Perception and Cognitive Implications of Logarithmic Scales for Exponentially Increasing Data: Perceptual Sensitivity Tested with Statistical Lineups

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Abstract

Logarithmic transformations are a standard solution to displaying data that span several magnitudes within a single graph. This paper investigates the impact of log scales on perceptual sensitivity through a visual inference experiment using statistical lineups. Our study evaluated participant's ability to detect differences between exponentially increasing data, characterized by varying levels of curvature, using both linear and logarithmic scales. Participants were presented with a series of plots and asked to identify the panel that appeared most different from the others. Due to the choice of scale altering the contextual appearance of the data, the results revealed slight perceptual advantages for both scales depending on the curvatures of the compared data. This study serves as the initial part of a three-paper series dedicated to understanding the perceptual and cognitive implications of using logarithmic scales for visualizing exponentially increasing data. These studies serve as an example of multi-modal graphical testing, examining different levels of engagement and interaction with graphics to establish nuanced and specific guidelines for graphical design.

Keywords: log scales, visual inference, graphical testing

1 Introduction

Effective communication of data is critical in influencing people's opinions and actions. This consideration was particularly true during the COVID-19 pandemic, where data visualizations and dashboards were vital in informing the public and policymakers about the outbreak's status. Local governments relied on graphics to inform their decisions about shutdowns and mask mandates, while residents were presented with data visualizations to encourage compliance with these regulations. A major issue designers encountered when creating COVID-19 plots was how to display data from a wide range of values (Fagen-Ulmschneider, 2020; Burn-Murdoch et al., 2020). When faced with data that span several orders of magnitude, we must decide whether to show the data on its original scale - where smaller magnitudes are compressed, making them harder to discern - or to transform the scale, which alters the contextual appearance of the data. Logarithmic axis transformations have emerged as a standard solution to this challenge, as they allow for the display of data over several orders of magnitude within a single graph.

Exponential data are one such example where, on a linear scale, the separation of data at smaller magnitudes becomes difficult to discern; Fig. 1 presents hard drive capacity over the past forty years on both the linear and log scales to demonstrate the usefulness of log scales when dealing with data spanning multiple magnitudes. Logarithms facilitate the conversion of multiplicative relationships (where the apparent distance between numbers like 1 & 10 or 10 & 100 grows multiplicatively) to additive relationships (displaying these same intervals as equal distances), highlighting proportional relationships and linearizing power functions (Menge et al., 2018). Logarithms also have practical applications, simplifying the computation of small numbers such as likelihoods and transforming data to conform to statistical assumptions. Although log scales have a long history of use in fields such

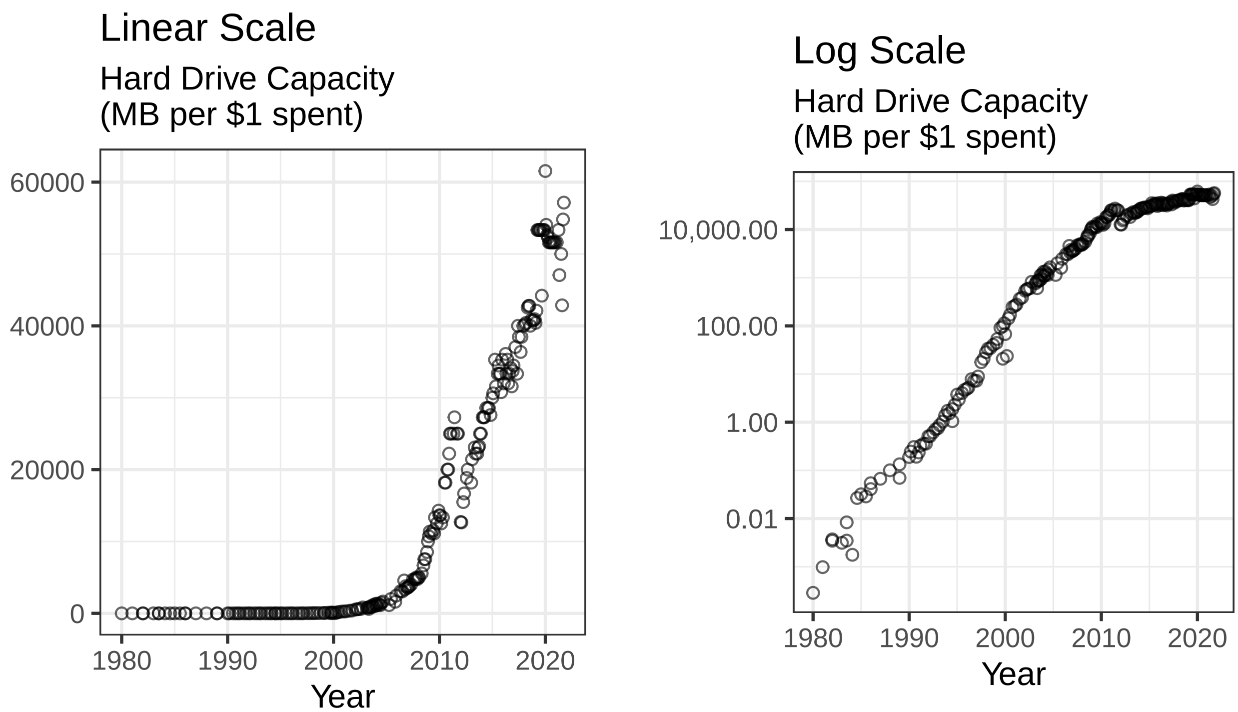


Figure 1: These plots present hard drive capacity (mem, 2021) over the past forty years on both the linear and log scale and illustrate the use of the log scale when displaying data which spans several magnitudes.

as ecology, psychophysics, engineering, and physics (Heckler et al., 2013; Waddell, 2005), there is still a need to understand the implications of their use and provide best practices for their implementation.

Apart from the biases resulting from using log scales, there is a general misinterpretation of exponential growth. Previous studies have explored the estimation and prediction of exponential growth and found that individuals often underestimate exponential growth when presented values numerically and graphically (Wagenaar and Sagaria, 1975; Ciccione et al., 2022). The hierarchy of plot objects, such as lengths and angles, as described by Cleveland and McGill (1985), offers a possible explanation for the underestimation observed in exponentially increasing trends. Experiments conducted by Wagenaar and Sagaria (1975), Jones (1977), and MacKinnon and Wearing (1991) aimed to improve estimation accuracy for exponential growth. While contextual knowledge or experience did not enhance estimation, instruction on exponential growth reduced underestimation by prompting participants to adjust their initial starting value (Wagenaar and Sagaria, 1975; Jones, 1977). Furthermore, providing immediate feedback to participants about the accuracy of their predictions improved estimation (MacKinnon and Wearing, 1991). Recently, Lammers et al. (2020) found that people misperceived exponential growth as increasing linearly, and that providing instructions to individuals about the exponential growth reduced this misperception. These misinterpretations and misperceptions can lead to decisions made under inaccurate understanding, resulting in potential consequences.

Log transforming the data may address our inability to predict exponential growth accurately. However, this transformation introduces new complexities, as most readers may need to be mathematically sophisticated enough to intuitively understand logarithmic math and translate it back into real-world effects. Despite the transformative power of logarithmic scales in facilitating accurate data representation, Menge et al. (2018)'s survey of ecologists highlights the challenges associated with the widespread comprehension of log-scaled data. Notably, the study identifies prevalent misconceptions arising from linear extrapolation assumptions in log-log space, a factor that often leads to neglect of the underlying exponential relationships in linear-linear space.

Building upon the need for a nuanced understanding of data representation, Buja et al. (2009) introduced statistical lineups as a framework for statistical inference and graphical tests. Statistical lineups treat a data plot as a visual statistic, summarizing the data as a numerical function or mapping. Evaluation of a panel in a statistical lineup requires visual inspection by a person, and if visual evaluations lead to different results, two visualization methods are deemed significantly different. Recent studies have utilized statistical lineups to quantify the perception of graphical design choices (Hofmann et al., 2012; Loy et al., 2017, 2016; VanderPlas and Hofmann, 2017). Statistical lineups provide an elegant way of combining perception and statistical hypothesis testing through graphical experiments (Majumder et al., 2013; Vanderplas et al., 2020; Wickham et al., 2010).

The term 'lineup' is an analogy to police lineups in criminal investigations, where witnesses identify the criminal from a group of individuals. Similarly, researchers present a statistical lineup plot consisting of smaller panels and ask the viewer to identify the panel that contains the actual data from a set of decoy null plots. Researchers generate null plots containing data generated according to a prespecified hypothesis using permutation or simulation. Typically, a statistical lineup consists of 20 panels, with one target panel and 19 null panels. If the viewer can identify the target panel from the null panels, it suggests that the actual data is visually distinct from the data generated under the null model.

While explicit graphical tests direct the participant to a specific feature of a plot to answer a particular question, implicit graphical tests require the user to identify both the purpose and function of the plot in order to evaluate the plots shown. Furthermore, implicit graphical tests, such as lineups, simultaneously test for multiple visual features, including outliers, clusters, and linear and nonlinear relationships (VanderPlas and Hofmann, 2015). Researchers can collect responses from multiple viewers using crowd-sourcing websites such as Prolific and Amazon Mechanical Turk.

In this paper, our primary focus is to evaluate the perceptual sensitivity towards the degree of curvature as a result of displaying data on linear and log scales. To address this, we conducted a visual inference experiment employing statistical lineups (Buja et al., 2009). Although our findings could have broad applications to various functions resulting in curvature, our experiment deliberately centered on participants' ability to identify differences in the curvature of exponentially increasing curves when presented on differing scales, thus altering their curvature appearance. We discuss the nuances and challenges of testing the perception of exponential growth in the appendix. Importantly, this investigation did not necessitate participants to undergo mathematical training or possess a prior understanding of exponential growth or logarithmic scales. Instead, it aimed to unravel the inherent ability to identify differences in curvature within charts, focusing on the fundamental nature of visual perception.

In Section 2 we describe the participant sample, the graphical task, data generation process, and study design. Section 3 describes the participant data collected and shares results from the statistical analyses of the data using a generalized linear mixed model. We present overall conclusions and discussion of the results in Section 4, and provide an overview of future related papers. The Supplementary Material includes a link to the RShiny data collection applet, participant data used for analysis, and code to replicate the analysis. The results of this study lay the groundwork for further exploration of the implications of using log scales in data visualization.

2 Study Development and Methods

2.1 Data Generation

In this study, we simulated data from a two-parameter exponential function with a vertical shift to generate the target and null data sets; the curve equations between panels differ in the parameter values selected for the null and target panels. In order to guarantee the simulated data spans the same domain and range of values for each statistical lineup panel, we began with a domain constraint of $x \in [0, 20]$ and a range constraint of $y \in [10, 100]$ with $N = 50$ points randomly assigned throughout the domain. We mapped the randomly generated x values to a corresponding y value based on the function curve with predetermined parameter values and multiplicative random errors to simulate the response. Finally, we applied a vertical shift constraint. These constraints assure that participants who select the target panel are doing so because of their visual perception differentiating between curvature or growth rate rather than different starting or ending values.

We simulated data based on a two-parameter exponential function extended with an additional parameter for multiplicative errors and a vertical shift, for a total of four parameters:

$$y_i = \alpha \cdot e^{\beta \cdot x_i + \epsilon_i} + \theta \tag{1}$$

$$\text{with } \epsilon_i \sim N(0, \sigma^2)$$

where α controls the vertical extent, β affects the rate of growth, and θ provides a vertical shift of the function. The parameters α and θ were necessary to adjust based on β and σ^2 to guarantee the range and domain constraints are met. It is worth noting that while the inclusion of θ allows for constraints on the range, when a log transformation is applied, the curve loses the linearity properties typically seen with log transformed exponential data.

Equation (1) was used to generate $N = 50$ points $(x_i, y_i), i = 1, ..., N$ where x and y have an increasing exponential relationship. We selected parameters and generated data using a heuristic approach rather than algebraic to ensure the parameter selections resulted in visually distinct curves.

In summary, we identified three points (start, midpoint, and end) we wanted the curve to pass through. These control the level of curvature seen in the data. Then we optimized a nonlinear least squares regression line using the function from Equation (1) to obtain parameters used for simulating the data sets with multiplicative errors and a vertical shift. We provide details for the heuristic data generation procedure in Algorithm 1 and Algorithm 2.

2.2 Parameter Selection

We chose three levels of trend curvature (low curvature, medium curvature, and high curvature). For each curvature level, we simulated 1,000 data sets of (x_{ij}, y_{ij}) points for $i = 1, ..., 50$ increments of x-values and replicated $j = 1, ..., 10$ corresponding y-values per x-value. Each generated x_i point from Algorithm 2 was replicated ten times. Using the `lm` function from base R, we fit a linear regression model on each of the individual data sets and included the explanatory variable as both numeric and categorical. We then computed the lack of fit statistic (LOF) which measures the deviation of the data from the linear regression model (Kutner et al., 2004; Stroup, 2013) using the `anova` function from base R to determine the F-test effect of the categorical explanatory variable. After obtaining the LOF statistic for each level of curvature, we evaluated the density plots (Fig. 2) to provide a metric for differentiating between the curvature levels and thus detecting the target plot. While the LOF statistic provides a numerical value for discriminating between the difficulty levels, it cannot be directly related to the perceptual discriminability; it serves primarily as an approximation to ensure that we are testing parameters at several distinct curvature levels. Table 1 lists the final parameters used for data simulation.

2.3 Lineup Setup

To generate the small multiple scatter plots for the statistical lineups shown to participants in the study, we simulated a single data set corresponding to curvature level A for the

Algorithm 1 Lineup Parameter Estimation

- **Input Parameters:** domain $x \in [0, 20]$, range $y \in [10, 100]$, midpoint x_{mid}.
- **Output Parameters:** estimated model parameters $\hat{\alpha}, \hat{\beta}, \hat{\theta}$.

1: In order to estimate three parameters, we jittered two x values around the pre-selected midpoint (depends on curvature) for a total of four points (x_{min}, $x_{mid} - 1$, $x_{mid} + 1$, x_{max})

2: The starting and ending x values correspond to y values from the predefined range providing two points, $(0, 0)$ and $(20, 100)$

3: To determine the y value corresponding to the midpoints, we first computed the point-slope form of the line passing through $(0, 100)$ and $(20, 0)$, resulting in $y = -5 \cdot x + 100$. Perceptually, when plotting with an aspect ratio of 1, this appears as a line with a $-45°$ angle, similar to what we typically see in a $y = -x$ line, although algebraically it is scaled to fit the assigned domain and range.

4: Next, we determined the y value associated with the two additional middle points, $x_{mid} - 0.1$ and $x_{mid} + 0.1$, using the equation of the line determined in (3). This provides us our four points for estimating the three parameters.

5: From the set of points (x_k, y_k) for $k = 1, 2, 3, 4$, calculate the coefficients from the linear regression model $\ln(y_k) = b_0 + b_1 x_k$ to obtain starting values for $\alpha_0 = e^{b_0}, \beta_0 = b_1, \theta_0 = 0.5 \cdot \min(y)$

6: Using the `nls` function from the base `stats` package in Rstudio (R Core Team, 2022) and the starting parameter values - $\alpha_0, \beta_0, \theta_0$ - fit the nonlinear model, $y_k = \alpha \cdot e^{\beta \cdot x_k} + \theta$ to get estimated parameter values for $\hat{\alpha}, \hat{\beta}, \hat{\theta}$.

Algorithm 2 Lineup Exponential Data Simulation

- **Input Parameters:** sample size $N = 50$, estimated parameters $\hat{\alpha}$, $\hat{\beta}$, and $\hat{\theta}$, from Algorithm 1, and standard deviation σ from the exponential curve.
- **Output Parameters:** N points, in the form of vectors $\mathbf{x}$ and $\mathbf{y}$.

1: Generate $\tilde{x}_j, j = 1, ..., \frac{3}{4}N$ as a sequence of evenly spaced points in $[0, 20]$. This ensures the full domain of x is used, fulfilling the constraints of spanning the same domain and range for each parameter combination.

2: Obtain $\tilde{x}_i, i = 1, ..., N$ by sampling $N = 50$ values from the set of $\tilde{x}_j$ values. This guarantees some variability and potential clustering in the exponential growth curve disrupting the perception due to continuity of points.

3: Obtain the final x_i values by jittering $\tilde{x}_i$.

4: Calculate $\tilde{\alpha} = \frac{\hat{\alpha}}{e^{\sigma^2/2}}$. This ensures that the range of simulated values for different standard deviation parameters has an equal expected value for a given rate of change due to the non-constant variance across the domain.

5: Generate $y_i = \tilde{\alpha} \cdot e^{\hat{\beta} x_i + e_i} + \hat{\theta}$ where $e_i \sim N(0, \sigma^2)$.

Table 1: Lineup data simulation final parameters

Curvature Level	x_{mid}	$\hat{\alpha}$	$\tilde{\alpha}$	$\hat{\beta}$	$\hat{\theta}$	$\hat{\sigma}$
High	14.5	0.91	0.88	0.23	9.10	0.25
Medium	13.0	6.86	6.82	0.13	3.14	0.12
Low	11.5	37.26	37.22	0.06	-27.26	0.05

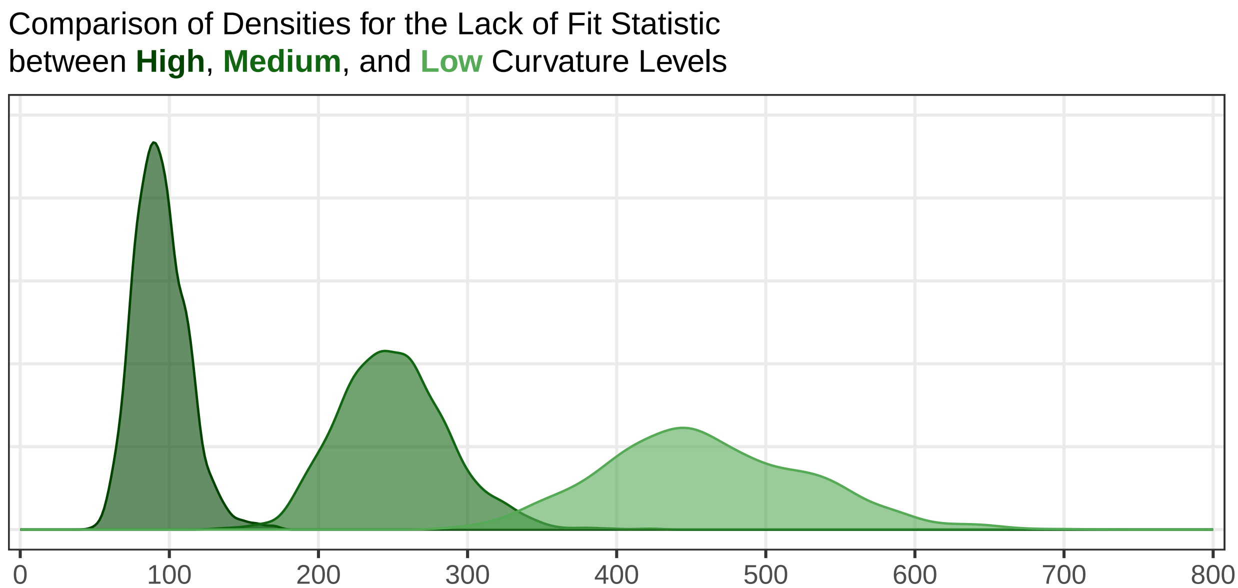


Figure 2: Density plot of the lack of fit statistic showing separation of difficulty levels: obvious curvature, noticable curvature, and almost linear.

target plot and multiple data sets corresponding to curvature level B for the null plots. The `nullabor` package in R (Buja et al., 2009) randomly assigned the target plot to one of the panels surrounded by panels containing null plots. The target and null panels span a similar domain and range due to the implemented constraints when simulating the data; the rationale for this decision is based on preattentive feature perception (Wolfe and Utochkin, 2019) and is discussed in detail in the appendix. There were a total of six lineup curvature combinations; Fig. 3 illustrates the six lineup curvature combinations (top: linear scale; bottom: log scale) where the solid line indicates the curvature level designated to the target plot while the dashed line indicates the curvature level assigned to the null plots. Two sets of each lineup curvature combination were simulated (a total of twelve test data sets) and plotted on both the linear scale and the log scale (24 test lineup plots). In addition to the experimental lineups, we generated two sets of homogeneous "Rorschach" lineups, each featuring panels simulated from the same distribution for each of the three curvatures (a

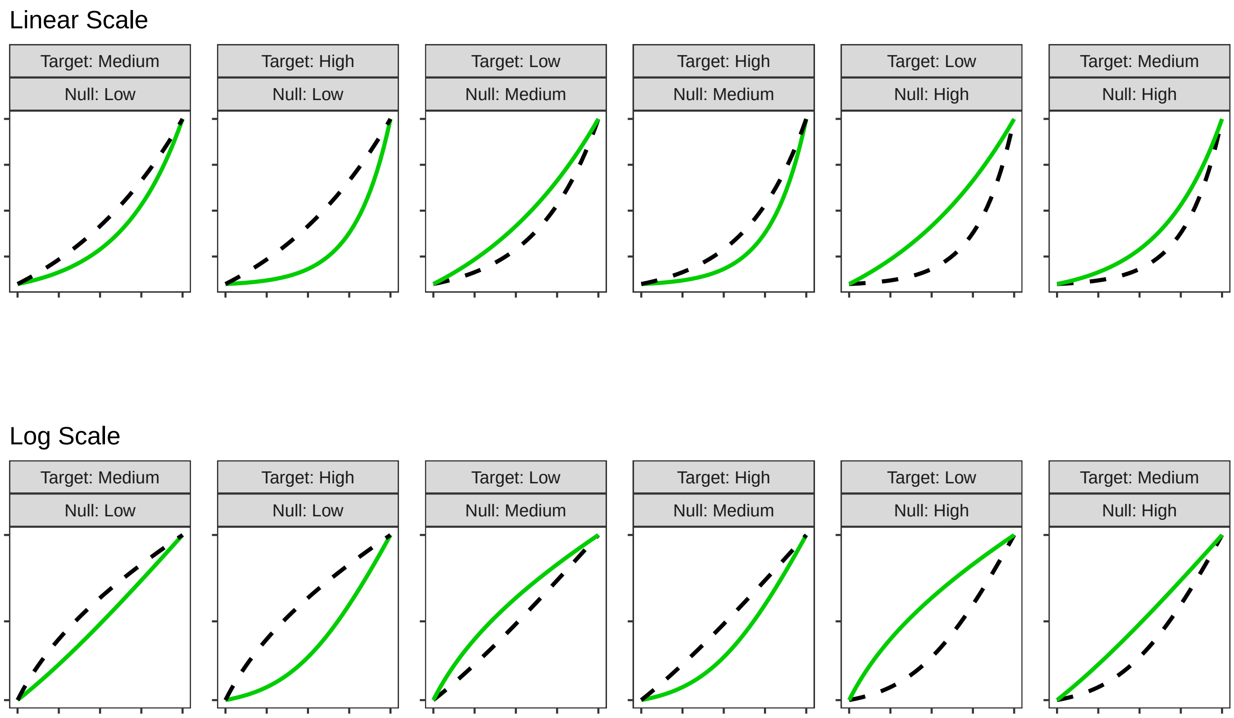


Figure 3: Thumbnail plots illustrating the six curvature combinations displayed on both scales (linear and log). The solid line indicates the curvature level to be identified as the target plot from amongst a set of null plots with the curvature level indicated by the dashed line.

total of six Rorschach data sets). Importantly, participants were unaware that there was no designated target panel for identification. Each participant evaluated one “Rorschach” lineup.

Fig. 4 presents examples of statistical lineups with the target data simulated with exponential parameters corresponding to high curvature and the surrounding null panels simulated with parameters for low curvature. The statistical lineup on the left presents increasing exponential data displayed on a linear scale with panel 13 as the target panel. The lineup on the right shows increasing exponential data plotted on a log scale with panel 4 as the target panel.

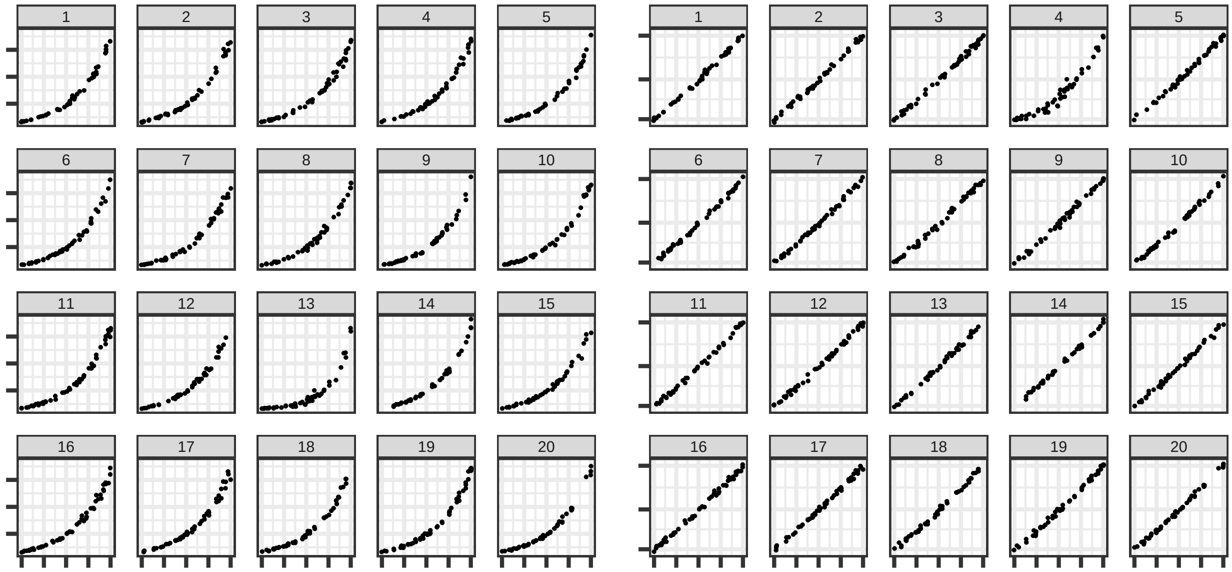


Figure 4: The lineup plot on the left displays increasing exponential data on a linear scale with panel 13 as the target. The lineup plot on the right displays increasing exponential data on the log scale with panel 4 as the target.

2.4 Study Design and Implementation

We used Prolific, a survey site that connects researchers to study participants, to recruit participants above the age of majority (18+ in most regions; 19+ in NE and AL; 21+ in MS) in their region. We did not request a representative sample, as previous literature indicates the presence of only minor effects of demographics on the outcome of graphical experiments involving lineups (VanderPlas and Hofmann, 2015; Majumder et al., 2013). Following the completion of the current statistical lineup study, participants sequentially completed two additional graphical experiments related to the perception of logarithmic scales (to be discussed at a later date), and we compensated them for their participation in the series of all three studies.

We showed each participant 13 lineup plots (12 test and one Rorschach). At the start of the study, we randomly assigned participants to one of the two replicate data sets for each of the six unique lineup curvature combinations. Participants evaluated the lineup plot corresponding to the linear and log scales for each assigned test data set. For the additional Rorschach lineup plot, we randomly assigned participants to one data set shown on either the linear or the log scale. We randomized the order in which participants saw the assigned 13 lineup plots.

Fig. 5 presents a screenshot of the study homepage, which served as an introduction to participants and guided them through the series of three graphical experiments. The first experiment, discussed in this paper, utilized statistical lineups to investigate the effect of logarithmic scales on perceptual sensitivity. The second experiment incorporated an interactive 'You Draw It' feature introduced by Aisch et al. (2015) and employed in the study by Robinson et al. (2022), to examine the effect of logarithmic scales on prediction. The third experiment focused on numerical estimation and cognitive understanding of logarithmic scales.

Participants completed the series of graphical tests using an R Shiny application (Chang et al., 2022) accessible at `https://emily-robinson.shinyapps.io/perception-of-statistical-graphics-log/`. The code used to implement the described methods for generating the data, setting up the lineups, and implementing the study design is available on GitHub at `https://github.com/earobinson95/perception-of-statistical-graphics-log`. Before completing the lineup study, the 'You Draw It' study, and the estimation study, participants were prompted to provide their demographic information. The series of studies received an IRB exemption (IRB #20200720178EX) from the University of Nebraska - Lincoln.

The statistical lineup study guided participants through the series of 13 lineup plots and asked them a single question per lineup, "Which plot is the most different?" (Fig. 6).

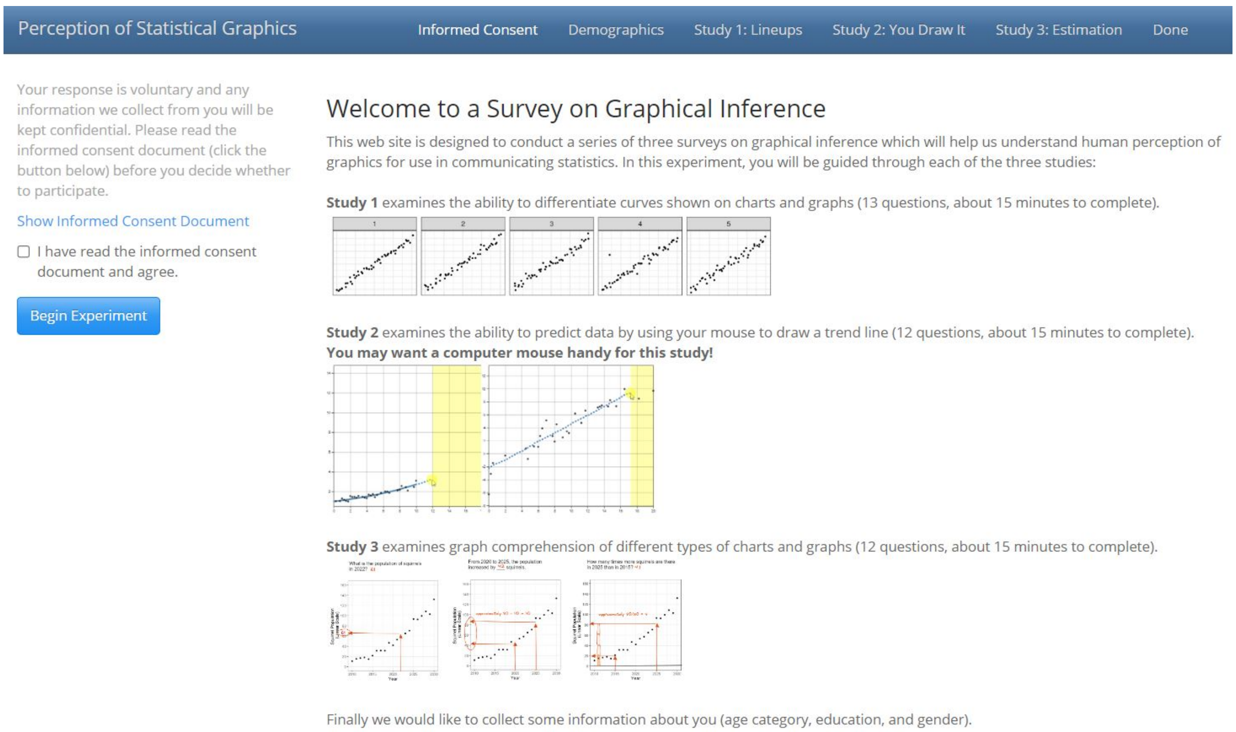


Figure 5: Screenshot of the study applet homepage guiding participants through the series of three graphical experiments related to the perception of logarithmic scales.

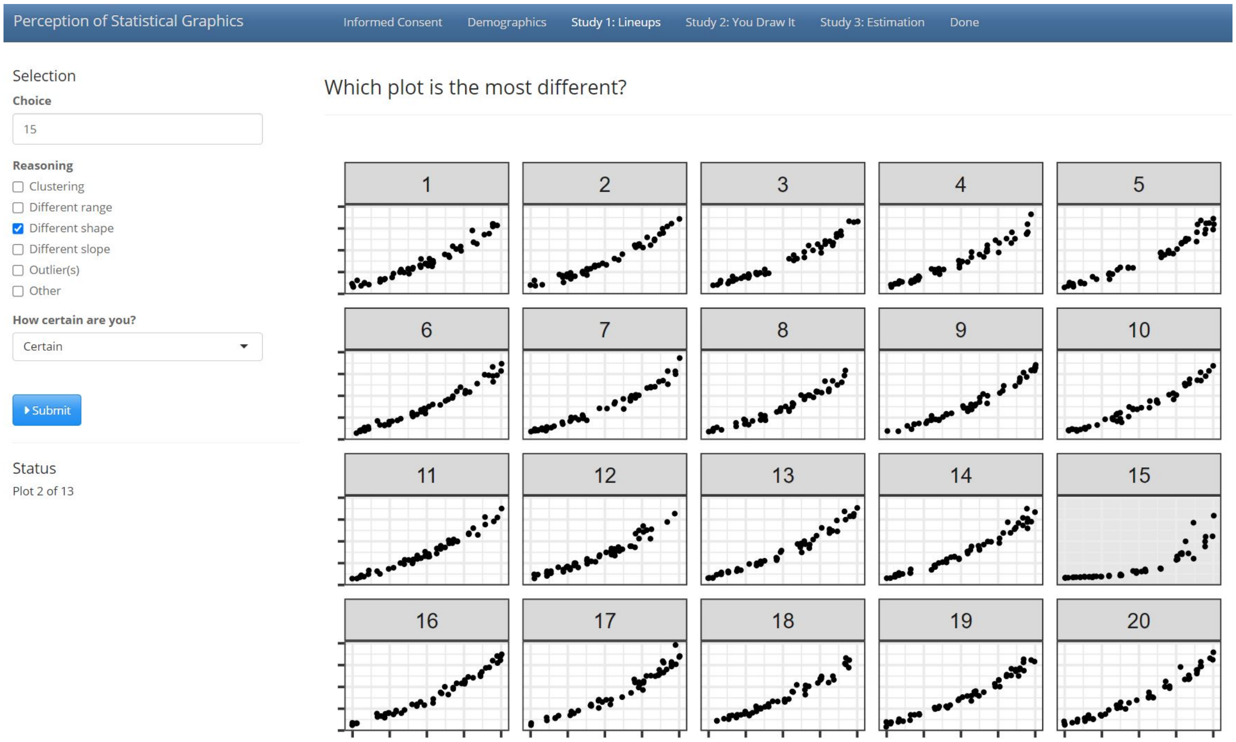


Figure 6: Screenshot of an example trial participants see when completing the lineup study. The applet guided participants through 13 lineup plots and asked them to identify the plot which appeared to be the most different from the others. After participants selected a panel, their choice was highlighted by gray (e.g., Panel 15).

In each lineup evaluation, participants then justified their choice in a select all that apply format (Clustering, Different range, Different shape, Outlier(s), and Other) under the "Reasoning" sidebar panel. Additionally, participants provided their level of confidence in their choice by answering "How certain are you?" on a five-point Likert scale from very certain to very uncertain. This graphical task aimed to test an individual's ability to perceptually differentiate exponentially increasing trends with differing levels of curvature on both the linear and log scales.

2.5 Statistical Analysis

In the first analysis, we aimed to understand our null data generation process. VanderPlas et al. (2021) provides an approach for calculating the statistical significance of lineups using a Dirichlet distribution. In particular, we are interested in calculating the parameter, α, of the Dirichlet distribution that best fits the "Rorschach" lineups. An $\alpha < 0.05$ indicates only one panel in the lineup was selected more frequently than others, whereas an $\alpha > 0.25$ provides support that participant selections are evenly divided across many interesting panels. We computed the α values using the `estimate_alpha_numeric` function from the vinference R package (Hofmann et al., 2020). This approach allows us to explore whether any null panels were significantly more unique, providing insights into the null plot sampling method.

To evaluate participant accuracy of test lineups, each lineup plot evaluated was assigned a binary value based on the participant response (correct target plot identification = 1, not correct target plot identification = 0). We defined Y_{ijkl} as the event that participant $l = 1, ..., N_{\text{participant}}$ correctly identified the target plot for data set $k = 1, 2$ with curvature combination $j = 1, 2, 3, 4, 5, 6$ plotted on scale $i = 1, 2$. The binary response was analyzed using a generalized linear mixed model (GLMM) following a binomial distribution with a logit link function with a row-column blocking design accounting for the variation due to participant and data set, respectively, as

$$\text{logit } P(Y_{ijkl}) = \eta + \delta_i + \gamma_j + (\delta\gamma)_{ij} + s_l + d_k \tag{2}$$

where

- η is the baseline average probability of selecting the target plot
- δ_i is the effect of scale $i = 1, 2$

- γ_j is the effect of curvature combination $j = 1, 2, 3, 4, 5, 6$

- $(\delta\gamma)_{ij}$ is the two-way interaction between the i^{th} scale and j^{th} curvature combination

- $s_l \sim N(0, \sigma^2_{\text{participant}})$ is the random effect for participant characteristics

- $d_k \sim N(0, \sigma^2_{\text{data}})$ is the random effect for data specific characteristics.

We assumed the random effects for data set and participant are independent. Target plot identification was analyzed using a GLMM implemented in `glmer` from the `lme4` R (version 4.2.2) package (Bates et al., 2015). We used the `emmeans` R package (Lenth, 2021) to calculate the estimated target detection probabilities and obtain odds ratio comparisons between the log and linear scale.

3 Results

We recruited participants and conducted the study via Prolific, a crowd-sourcing website, in March 2022. The study included a diverse group of participants, with an inner-quartile age range between 23 and 31 and a median age of 26 years old. Among the participants, 59% self-identified as male, 40% as female, and 1% as variant/nonconforming. Additionally, individuals from more than 21 countries participated in the study and 97% of the participants indicated fluency in English. Moreover, 81% of the participants reported having completed some undergraduate courses or higher.

3.1 Rorschach Evaluations

Fig. 7 illustrates the selection proportions for each panel on the "Rorschach" lineups corresponding to different curvature combinations and replicated data sets, when presented on both log and linear scales. Notably, the panel number is irrelevant, and the darker shade (bottom) denotes panels chosen more frequently than others. The plots exhibit relatively even distributions, offering multiple candidates for the most interesting panel. Additionally, most α values are calculated to be above 0.25, indicating participant attention is divided across multiple interesting panels. It is worth noting that in the low curvature "Rorschach" lineups simulated in data set replication one, a single panel stands out, with over 50% of participants selecting the same null panel on the linear scale and just under 50% on the log scale. This particular lineup yields an $\alpha = 0.17$ indicating that while the panel selection is not evenly distributed, there are more than one interesting panel. It is also worth noting that set two of the low curvature lineup results in a larger $\alpha = 0.4$. High α values, such as $\alpha = 3.72$ observed in the low curvature set 2 lineup, reflect a relatively uniform distribution of participant selections, indicating that multiple panels attracted similar levels of attention. This suggests the appropriateness of our null sampling method in producing visually diverse lineups, as supported by a visual examination of "Rorschach" evaluations and the calculation of α parameters.

3.2 Participant Accuracy

During data collection, 325 individuals completed 4,492 individual test lineup evaluations. We included only participants in the final analysis who completed the entire study, which included 311 participants and 3,958 lineup evaluations. Due to server capacity, some participants were required to restart the study, thus resulting in the possibility of more than twelve lineup evaluations per participant. As a whole, participants evaluated each uniquely generated lineup plot between 141 and 203 times (Mean: 164.92, SD: 14.9). Participants correctly identified the target panel in 47% of the 1,981 lineup evaluations made on the

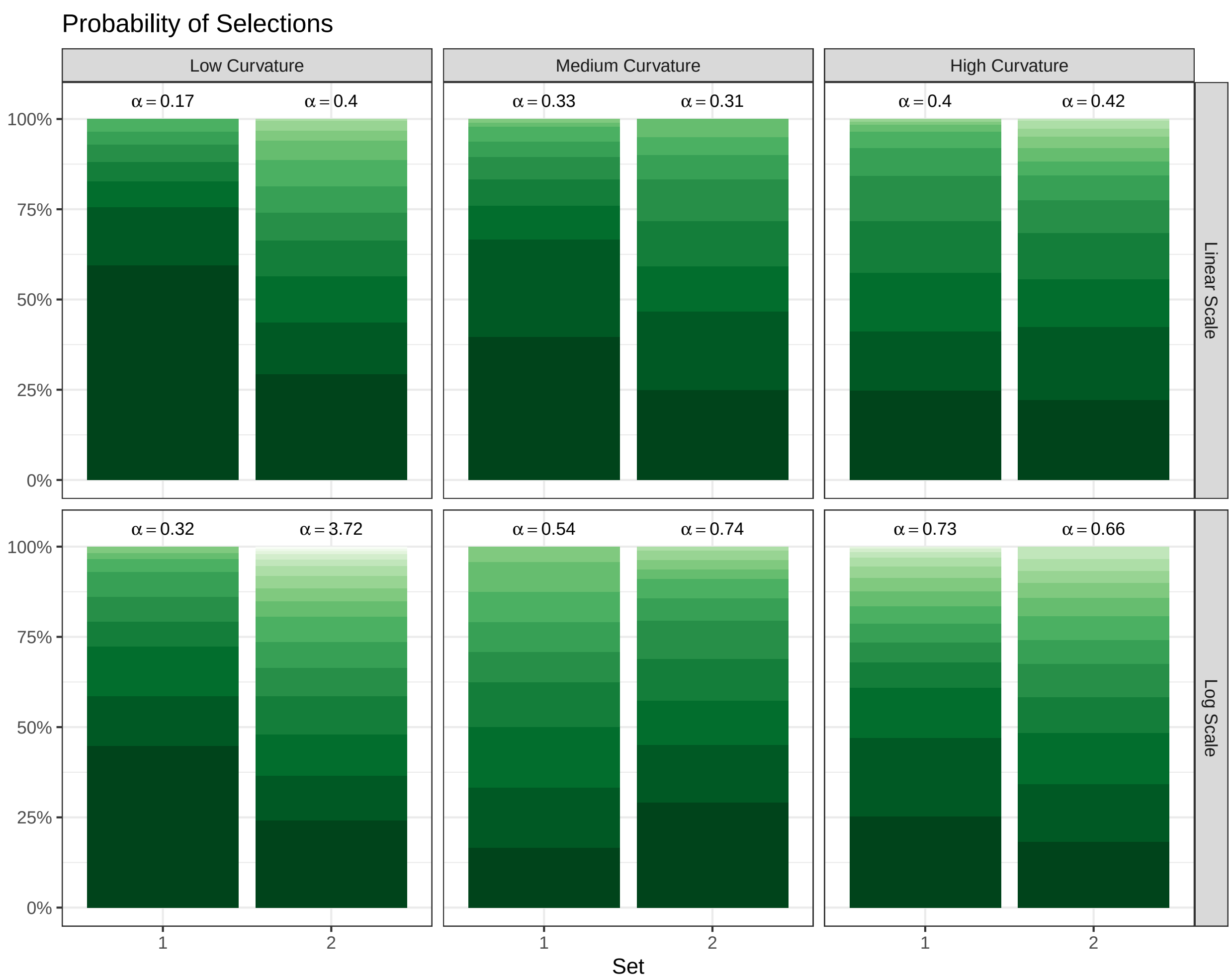


Figure 7: Observed probabilities of panel selections for the 'Rorschach' lineups. The darker shade (bottom) denotes panels chosen more frequently than others. The annotated text provides the calculated α parameter of the Dirichlet distribution that represents individuals' selections for the Rorschach plots.

linear scale and 65.3% of the 1,977 lineup evaluations made on the log scale. Fig. 8 shows the observed participant accuracy for each scale and curvature combination scenario with the dotted line indicating the average selection percentage for the most picked panel in the Rorschach lineups for each null curvature model. For example, on average, we found the medium curvature target model embedded in a background of low curvature null models have an accuracy rate of 76% on the linear scale while the associated top panel selected for low curvature Rorschach lineups were selected 47.6% of the time, on average. We can see from the observed results that participant accuracy for the linear scale ranged from 3.3% to 91.6% while participant accuracy when identified on the log scale ranges from 46.6% to 89%. On both the log and linear scales, the highest accuracy occurred in lineup plots where the target model and null model had a considerable difference in curvature (low curvature target plot embedded in high curvature null plots; high curvature target plot embedded in low curvature null plots), and the target plot had more curvature than the null plots (high curvature target plot embedded in low curvature null background plots). There was a decrease in accuracy on the linear scale when comparing a target plot with less curvature to null plots with more curvature (medium curvature target plot embedded in high curvature null plots; low curvature target plot embedded in medium curvature null plots; low curvature target plot embedded in high curvature null plots). Additionally, accuracy increased when data was displayed on the log scale compared to the linear scale in all curvature scenarios with an exception of a medium target curve embedded in low null curves. The dotted lines in Fig. 8 indicate that the average selection percentage for the top panels in the Rorschach lineups range from 22.8% to 47.6%. Given all panels in the Rorschach lineups were generated under the same model, this random selection highlights the variability in null plots due to random data generation. When pairing the accuracy with the selection percentage for the top picked panel from the Rorschach lineups, we can see that accuracy is higher than the randomly selected top Rorschach panel when presented on the log scale for all curvature combinations. However, when comparing accuracy to the top picked panel from the Rorschach lineups on the linear scale, accuracy of selections for low target panels when embedded in medium curvature background null panels and medium target panels embedded in high curvature background null panels is no better than the selection percentage for the top Rorschach panels associated with the null models; in these two cases, the variability in the null plots overpowers the actual trend differences. In other words, the linear scale appears to amplify the differences that are spurious due to random noise, such as outliers, in addition to amplifying the data trend. In contrast, the log scale minimizes the spurious differences due to random noise in favor of the differences due to trends in the data. As we replicated each curvature combination and Rorschach lineups with two sets in order to understand the variability, we can (minimally) examine both due to the data mean and variability in plot selections due to the data generation process. The appendix includes the number of selections for lineup plots from the study where the variability dominates the curvature trends found in the data.

In addition to participant accuracy, we conducted a linear mixed model on participant confidence (5 - Very Certain to 1 - Very Uncertain). We found that, on average, individuals who were correct are 0.556 (se 0.039) Likert scale points more confident than individuals who were incorrect in their plot choice across all conditions ($F = 214.57$; $df = 1, 3709$; $p < 0.0001$). Similarly, on average, lineups evaluated on the log scale indicated participants were only 0.075 (se 0.0366) Likert scale points more confident than lineups evaluated on the linear scale ($F = 3.70$; $df = 1, 3709$; $p = 0.054$). We discuss further details regarding selection reasoning and confidence level in the appendix.

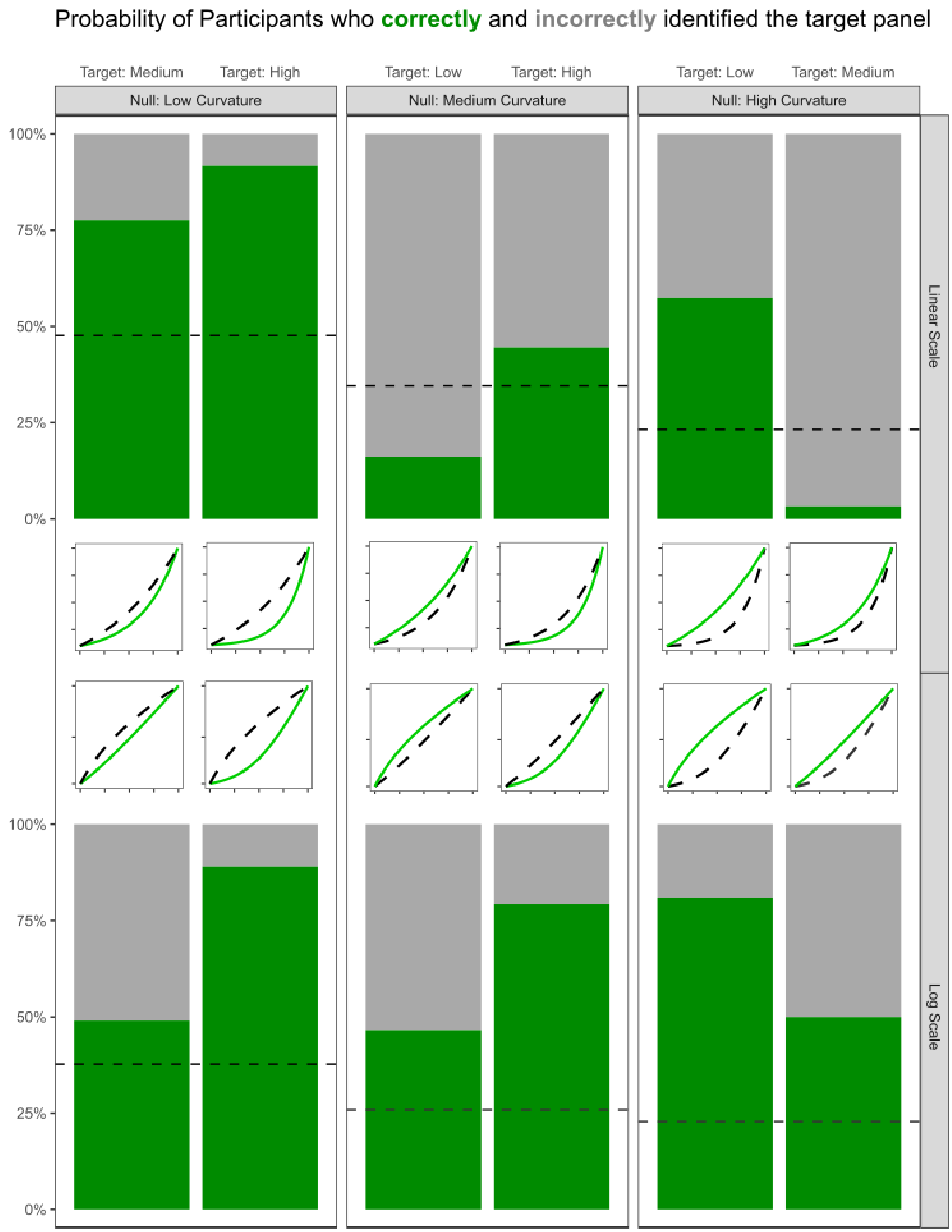


Figure 8: The observed accuracy for identifying the target panel for each curvature combination scenario with the accuracy for the linear scale shown on the top and the accuracy for the log scale shown on the bottom. The thumbnail figures below each plot display the

The results from the GLMM on participant accuracy indicated a strong interaction between the curvature combination and scale ($\chi^2_5 = 294.443$; p < 0.0001), and the estimated variance due to participant and data set were $\hat{\sigma}^2_{\text{participant}} = 1.19$ (s.e. $= 1.09$) and $\hat{\sigma}^2_{\text{data}} = 0.433$ (s.e. $= 0.66$), respectively. Therefore, we concluded that there was low variability in the accuracy of target panel detection between participants and across replications of uniquely simulated data sets. To determine the effect of scale, we compared the estimated accuracy between the log and linear scale within each curvature combination due to the interaction as determined by the GLMM.

Fig. 9 displays the estimated (log) odds ratio of successfully identifying the target panel on the log scale compared to the linear scale. The thumbnail figures to the right of the plot illustrate the curvature combination on the linear (left thumbnail) and log base ten (right thumbnail) scales associated with the y-axis label. The choice of scale had no impact if the curvature differences were substantial and the target plot had more curvature than the null plots (high curvature target plot embedded in low curvature null plots). However, presenting data on the log scale makes us more sensitive to slight changes in curvature (low or high curvature target plot embedded in medium curvature null plots; medium curvature target plot embedded in high curvature null plots) and apparent differences in curvature when the target plot had less curvature than the null plots (low curvature target plot embedded in high curvature null plots). An exception occurred when identifying a plot with curvature embedded in null plots close to a linear trend (medium curvature target panel embedded in low curvature null panels). The results indicate that participants were more accurate at detecting the target panel on the linear scale than on the log scale. When examining this curvature combination, the same perceptual effect occurred as we previously saw, but in a different context of scales. On the linear scale, participants were

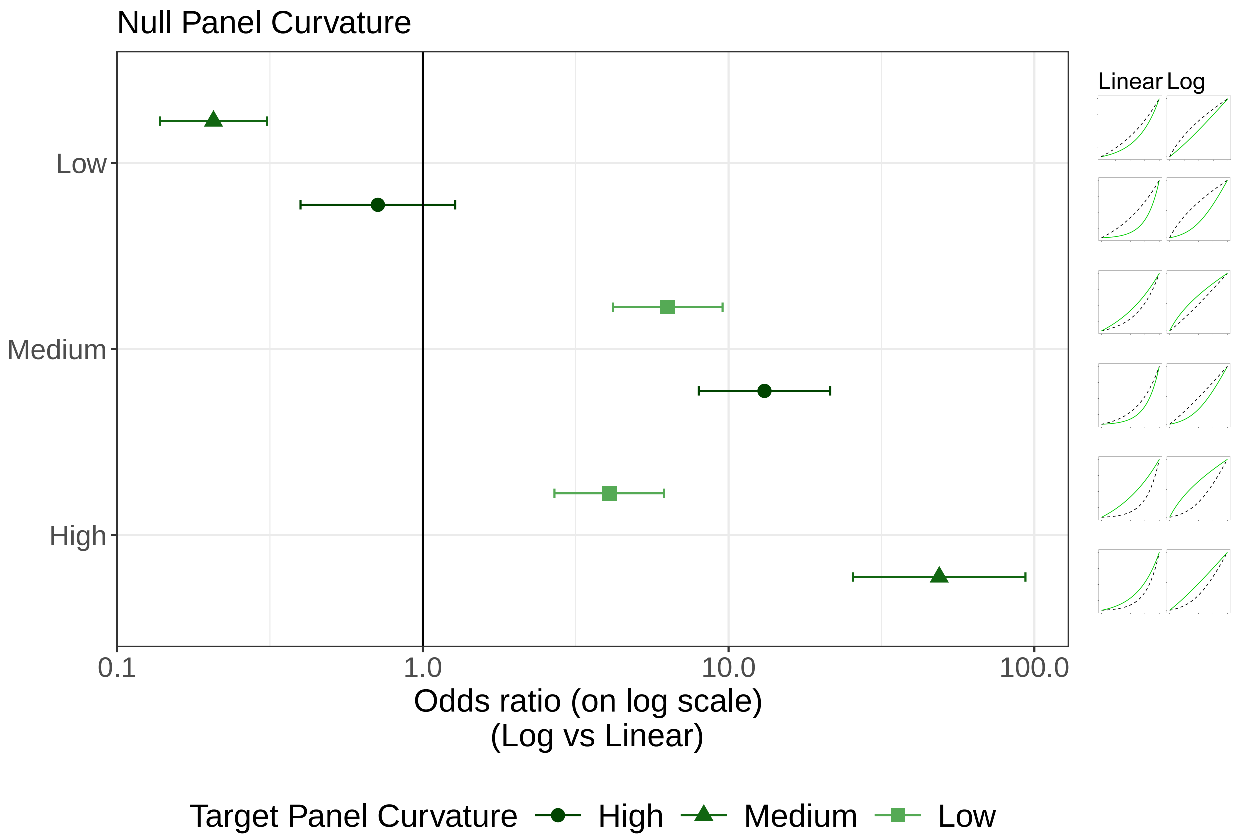


Figure 9: Estimated (log) odds ratio of successfully identifying the target panel on the log scale compared to the linear scale. The y-axis indicates the model parameters used to simulate the null plots with the target plot model parameter selection designated by shape and shade. The thumbnail figures on the right display the curvature combination as shown in Fig. 3 on both scales (linear - left, log - right).

perceptually identifying a convexly curved trend from close to a linear trend, whereas after the logarithmic transformation, participants were perceptually identifying a trend close to linear from a concavely curved trend (Fig. 3).

4 Discussion and Conclusion

This work aims to provide foundational research to support the principles that guide design decisions in scientific visualizations. In this study, we explored the perceptual sensitivities of detecting differences between curvature levels as altered by the choice of scale.

The results indicated that when there was a considerable difference in curvature between the target plot and null plots and the target plot had more curvature than the null plots, the choice of scale had no impact, and participants accurately differentiated between the two curves on both the linear and log scale. However, due to the altered appearance of the data, displaying exponentially increasing data on a log scale improved the accuracy of differentiating between models with slight curvature differences or apparent curvature differences when the target plot had less curvature than the null plots. An exception occurred when identifying a plot with curvature embedded in surrounding plots perceived close to a linear trend. The thumbnail figures in Fig. 8 visually demonstrate the opposing perceptual behaviors of the curves for a medium target curve embedded in low null curves displayed on the two different scales. This result supports the findings by Best et al. (2007), who found that the accuracy of identifying the correct curve type was higher when presented with nonlinear trends, indicating that it is hard to say something is linear (i.e., something has less curvature), but easy to say that it is not linear; our results concur with this observation. Therefore, in the context of this study, it is easy to identify a curve in a group of lines but harder to identify a line in a group of curves.

Using visual inference to identify these guidelines suggests that it is important to consider the appearance of curves as altered by the choice of scale when identifying differences between data. In the specific case of this study and selected data simulation equations, we found a slight advantage of using the log transformation when curvature differences were small due to the way the log scale changed the appearance of the curves. This aligns with findings by Dresp-Langley (2015), who demonstrated that the human visual system has an in-built sensitivity to local curve geometry, indicating that the visual perception of curvature can be significantly influenced by geometric transformations such as log scaling. The most frequent panel selection percentage from the Rorschach lineups indicates the linear scale highlights random variability in favor of the curvature trend while the log scale suppresses the spurious differences in favor of the actual trend differences in data.

It is worth noting that our study focused specifically on data simulated with a two-parameter exponential function with a shift and such conclusions may not be broadly applicable to other functions or different parameterizations of exponential data resulting in curvature. This scenario, while fairly specific, lays the perceptual groundwork for more investigation into the use of log scales with exponential data. Now that we know how curvature can be distinguished, it's easier to conduct follow up studies that cover more scenarios and use different graphical testing methods.

We conducted this study as the first in a series of three graphical tests to understand the perceptual and cognitive implications of using log scales to display exponentially increasing data. Previous research by Ciccione et al. (2022) found that individuals often underestimate exponential growth when presented values numerically and graphically, highlighting the need for better understanding and presentation of such data. In our next two papers in this series, we will investigate whether using log scales presents cognitive disadvantages, such as making it harder to utilize graphical information. Lammers et al. (2020) showed that correcting exponential growth bias could significantly increase support for social distancing, suggesting that improving the accuracy of visual representations of exponential growth can have substantial real-world impacts.

These studies serve as an example of multi-modal graphical testing, examining different levels of engagement and interaction with graphics in order to produce nuanced, specific guidelines for graphical design. By testing graphics in situations similar to how they are used in practice, we can gain additional insight into graphical perception and improve visual communication of scientific results.

Supplementary Material

- **Participant Data:** De-identified participant data collected in the study and used for analyses (lineup-model-data.csv).
- **Data Analysis Code:** The code used to replicate the analysis in this paper (lineups-analysis.qmd).
- **Study Applet Code:** The code used to create the study applet via RShiny can be found on GitHub at `https://github.com/earobinson95/perception-of-statistical-graphics-log`
- **README:** File containing detailed descriptions of the supplementary material. (README.html).

A Exponential growth rates and curvature

Let us consider a simple equation, distinct from the one employed to simulate data in our study. This equation can be used to simulate data which grows exponentially in x, with optional error term ϵ:

$$y = \exp\{\beta_1 x + \epsilon\} \quad \text{where} \quad \epsilon \sim N(0, \sigma). \tag{A1}$$

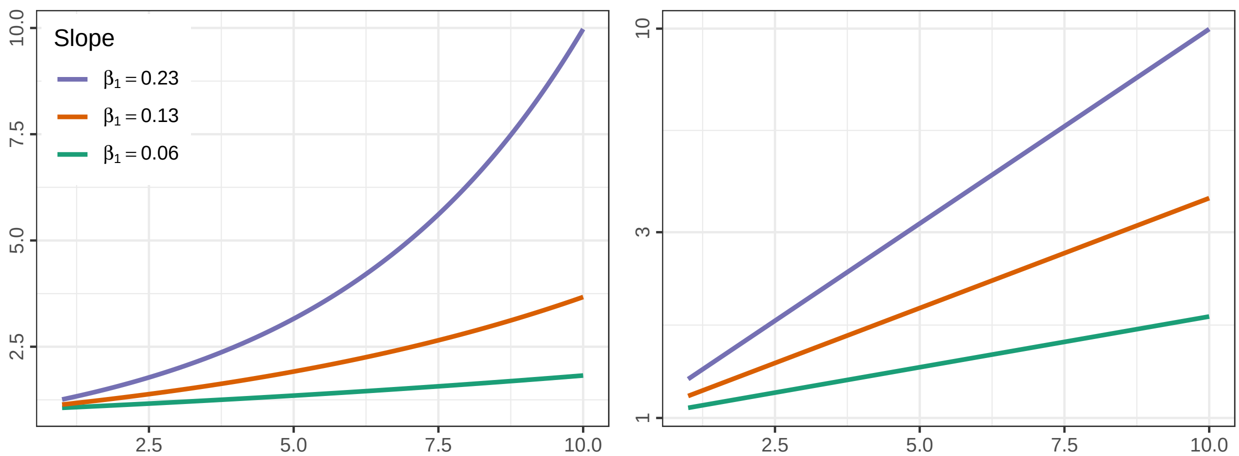


Figure A1: Linear and log scale simple exponential equations as specified in Eq. (A1), with error term excluded. The y-axis endpoints are a primary visual signal when differentiating between the lines in both plots.

It is important in lineup studies to control for extraneous visual signals, and when other visual signals creep in, results of the study can be difficult to interpret (VanderPlas and Hofmann, 2017).

In this particular case, the extraneous visual signals are the endpoints of the lines (the endpoint at $x = 10$ on a linear scale, and both endpoints on the log scale), as shown in Fig. A1. Preattentive features guide attention (Wolfe and Utochkin, 2019); one of the first things we do when scanning a graph is to notice the extent of the data within the axes. We must control the endpoints to ensure that participants are using active attention to assess

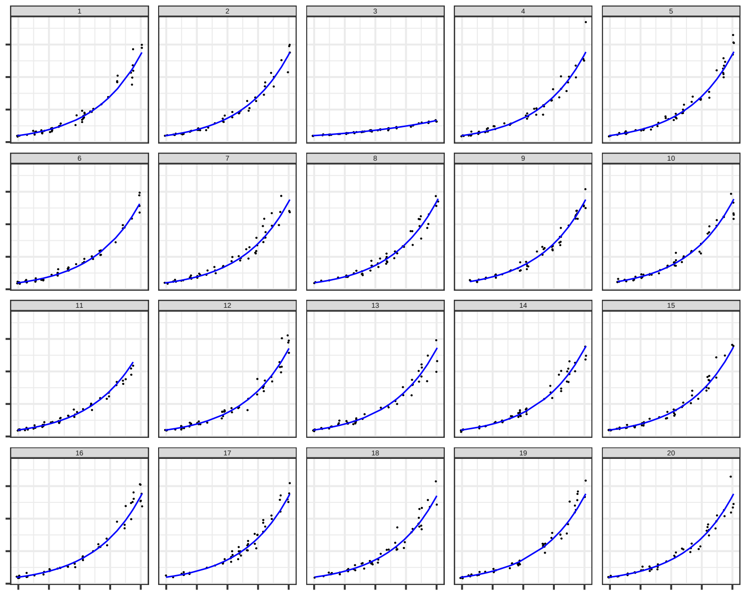


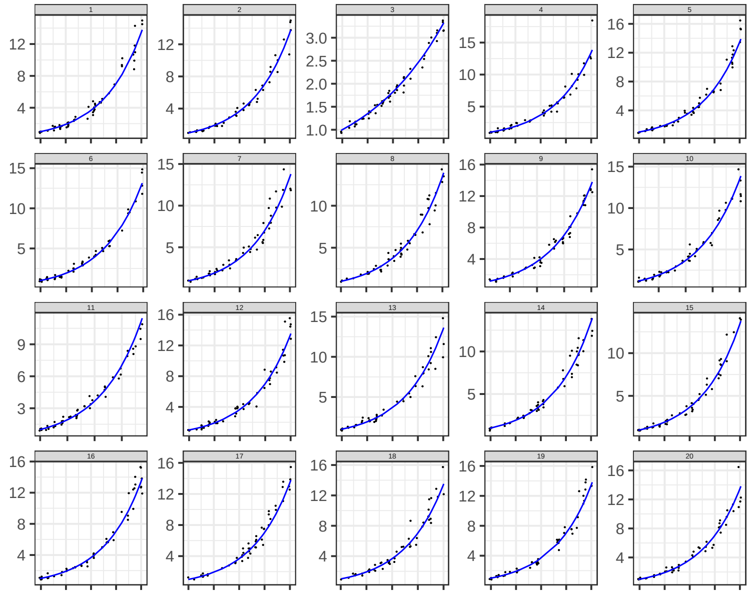
Figure A2: When axis limits are fixed to the most expansive extent in the generated data, the primary signal becomes the extent of the domain which has data, rather than the growth rate itself.

the lineup and draw conclusions, so that the whole visual signal is processed as part of the lineup evaluation; if participants are making decisions off of the endpoints, then we cannot interpret the results to say that they are assessing the rate of exponential growth rather than the end result.

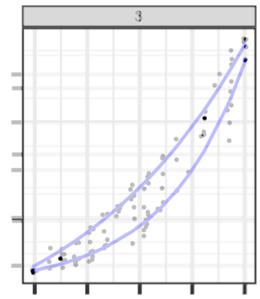
There are several different options for controlling the visual signal in a lineup:

0. Do nothing and set each subplot's limits to the overall maximum limits.
1. allow each subplot to have different axis limits
2. truncate the displayed range of each subplot to the minimum range generated in the lineup
3. add extra parameters to the exponential equation to adjust the range of the data so that it fits within a pre-specified domain, as in Algorithm 2.

Typically, lineups do not show axis labels or scaling, because the goal is to assess the



(a) Lineup



(b) Overlay of null and signal panels.

Figure A3: When y-axis limits vary by panel, the y-axis labels are the primary visual signal; the secondary signal (when the panels are overlaid) is the curvature of the line.

signal from the data without additional context. In addition, the traditional 20-panel lineup would be quite visually confusing in this circumstance:

Fig. A3 demonstrates that when we ignore the y-axis labels and overlay the signal panel with one of the null panels, the remaining difference is the curvature of the line. If we want to keep the standard convention of not including y-axis labels in lineups for simplicity and to reduce plot clutter, this alternative signal seems promising (and as we will show, is approximately equivalent to option 3).

Let us next consider option 2: Crop each panel to the minimum limits of all generated data. The result is shown in Fig. A4. This operation shifts which axis becomes the visually important factor, but doesn't change the problem: previously the issue was the y-axis extent, now it is the x-axis extent. Both of these parameters are implicitly affected by how we generate exponential data. In order to truly assess whether people can discriminate

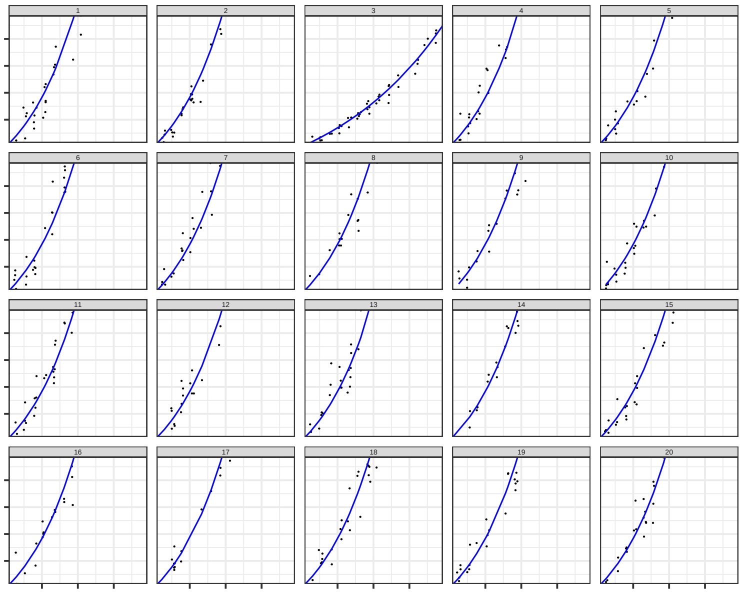


Figure A4: When axis limits are fixed to the most restrictive in all generated data, the primary signal becomes the extent of the x axis which is covered with data, rather than the growth rate itself.

between comparable exponential growth rates graphically, it is more useful to approach the problem from a curvature perspective rather than to artificially limit the data shown.

This issue is a fundamental problem when testing graphics: the test must meet the "goldilocks" standard - not too hard, not too easy, but just right. Both option 0 and option 2 fail this standard.

Visually, both the x and y axes matter equally, even if it might on paper make sense to truncate x in order to control the y axis range.

An additional benefit of controlling endpoints is that it also provides some realism: our goal was to assess the ability to examine exponential growth **rates**, motivated by the COVID-19 pandemic. As each geographic reporting region started with different numbers of cases and had different control policies, the α and θ parameters are reasonable to represent some of these differences while still examining the underlying exponential behavior.

To control these endpoints on both scales then we have to move to an exponential model that is a bit more complicated:

$$y = \alpha \cdot \exp\{\beta_1 x + \epsilon\} + \theta \quad \text{where} \quad \epsilon \sim N(0, \sigma). \tag{A2}$$

α and θ were used to make endpoints consistent among the growth rates. As a result, the lineups examine the degree of inflection of the trend rather than the explicit exponential growth rate. An explanation for how the ultimate values for α and θ were determined is provided in Algorithm 2. Fig. A5 uses the parameters in Table 1 and the data generating method described in Algorithm 2. The result is a series of panels which have obvious variation due to the points (a desirable feature) but where the underlying relationship shown in blue has consistent endpoints. As a result, the question becomes whether participants can identify that plot 3 has a different growth rate (as measured by the curvature of the line).

B Participant confidence and accuracy

In addition to investigating how scale and curvature rate influence participant accuracy, we asked participants to indicate their level of confidence on a five-point Likert scale by answering, "How certain are you?" Fig. B6 shows the observed confidence rating proportions for correct and incorrect identifications across the different scale and curvature combination scenarios. We conducted a linear mixed model using the lmer function from the lme4 (Bates et al., 2015) R package in order to determine how participant confidence (5 - Very Certain to 1 - Very Uncertain) varied across conditions. We included curvature combination, scale, accuracy (correct or incorrect), and all two- and three-way interactions along with a random participant effect. While there are limitations to assuming normality

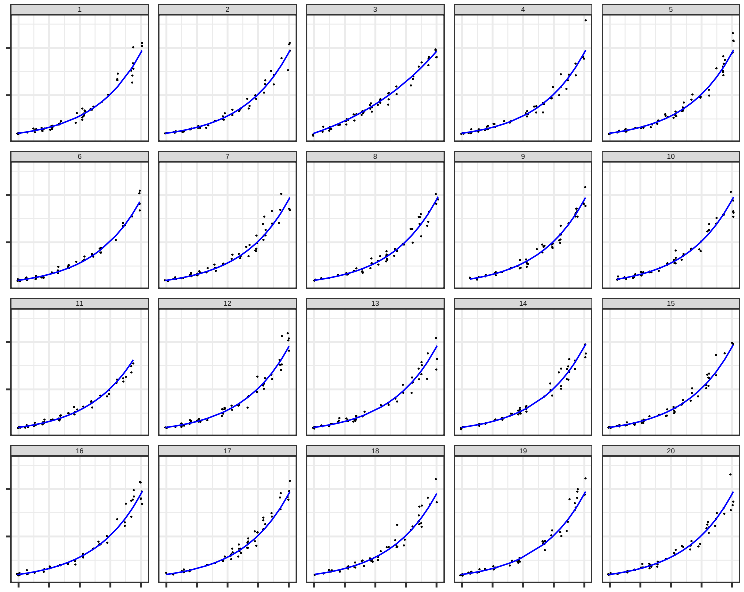


Figure A5: When the data are scaled such that endpoints are similar across all conditions, the primary feature becomes the curvature of the line, while individual plots show random variation due to the scatter around the line.

of Likert scale data, we evaluated model fit by observing residuals and comparing the results to non-parametric and multinomial methods. Due to the interpretability of the linear mixed model, we proceed to present the findings from this analysis. Overall, on average, individuals who were correct are 0.556 (se 0.039) Likert scale points more confident than individuals who were incorrect in their plot choice across all conditions (F = 214.57; df = 1, 3709; $p < 0.0001$). We did not find convincing evidence of a discernable difference in confidence levels between scales (F = 3.70; df = 1, 3709; p = 0.054). Overall, on average, lineups evaluated on the log scale indicated participants were 0.075 (se 0.0366) Likert scale points more confident than lineups evaluated on the linear scale.

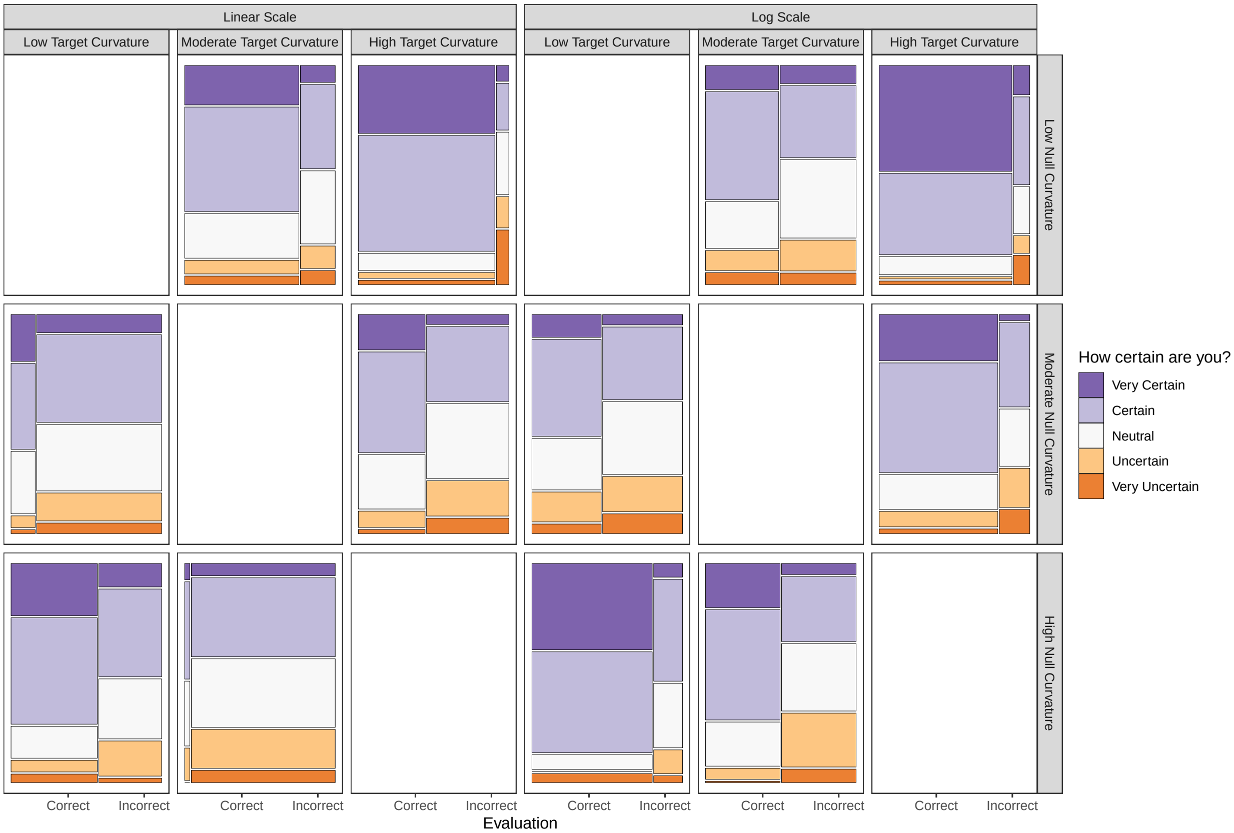
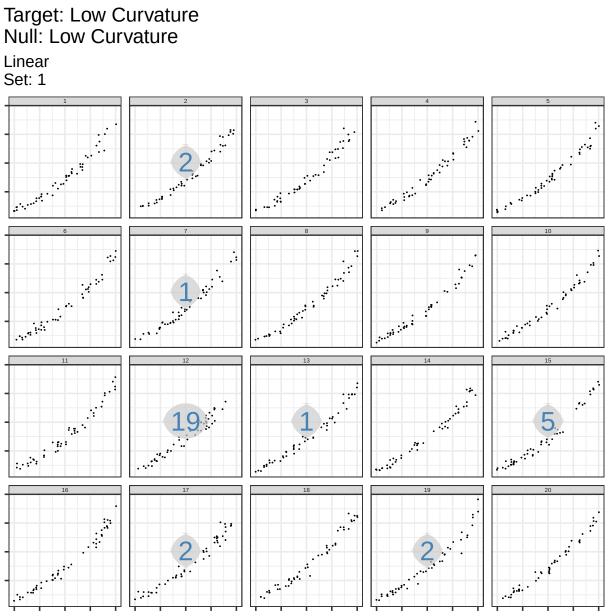


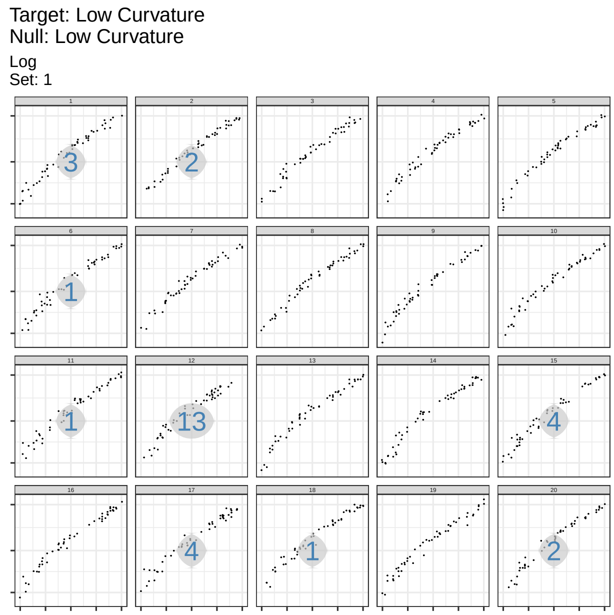
Figure B6: Observed proportion of evaluations for each confidence rating (five-point Likert scale) by accuracy. The width of the bars indicates the number of evaluations for correct and incorrect identifications within each curvature combination.

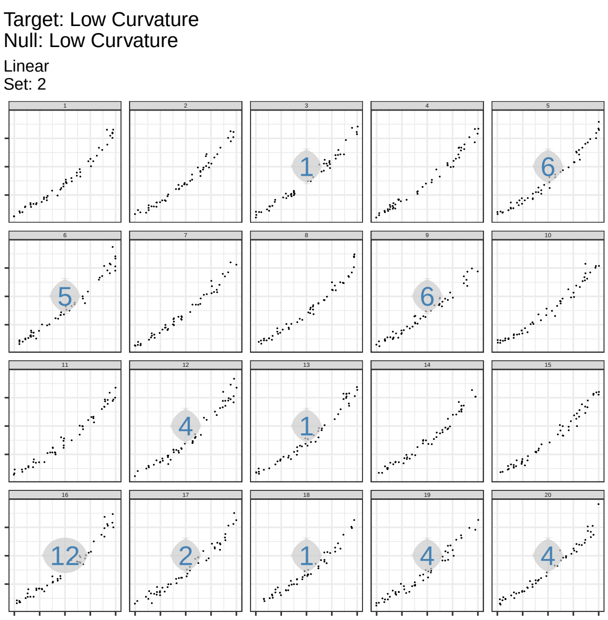
C Participant selections of lineup panels

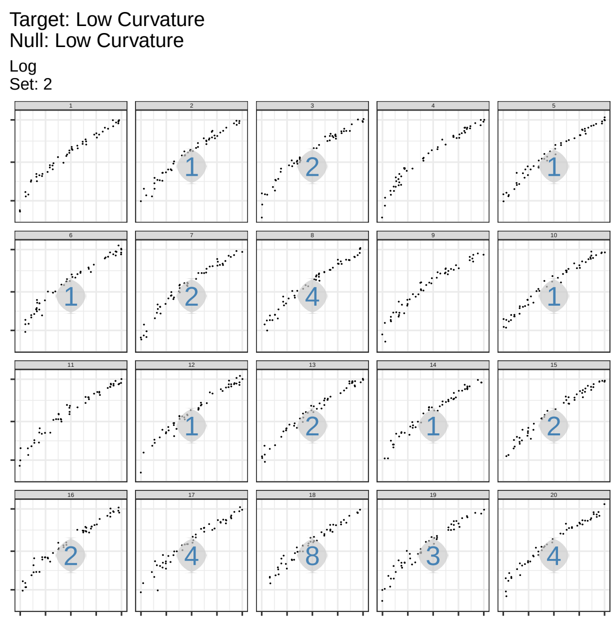
We simulated a total of 18 data sets (twelve test data sets and six Rorschach data sets) and plotted on both the linear scale and the log scale (36 lineup plots). Each parameter combination was generated in sets of two in order to account for variability in simulating data with random errors. The following plots display each of the lineups created for the study for the associated null background curvature model with the number of evaluation panel selection counts overlaid (circle = null panels; rectangle = target panel). In particular, set 2 for a high curvature target panel embedded in medium curvature background null panels highlights the outliers in the target plot (panel 14) on the linear scale while the log scale for the same data set suppresses the visual display of the outliers. By inspecting the lineups alongside the number of evaluations per panel, we can further understand whether participant's made their selection based on spurious differences due to random noise or the curvature trend of the data.

C.1 Low Curvature Null Background



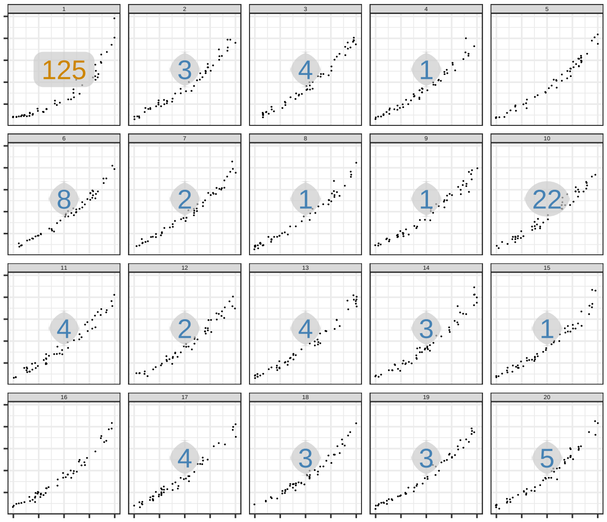






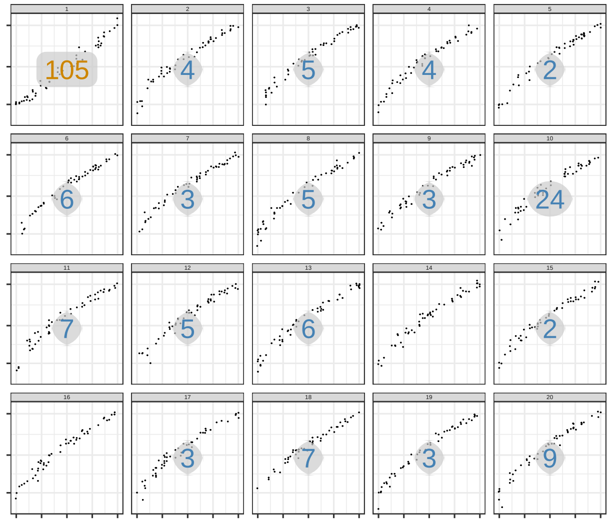
Target: Medium Curvature
Null: Low Curvature

Linear
Set: 1



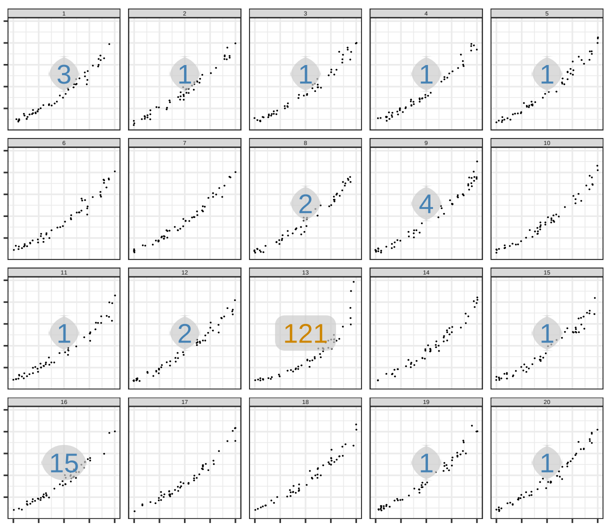
Target: Medium Curvature
Null: Low Curvature

Log
Set: 1



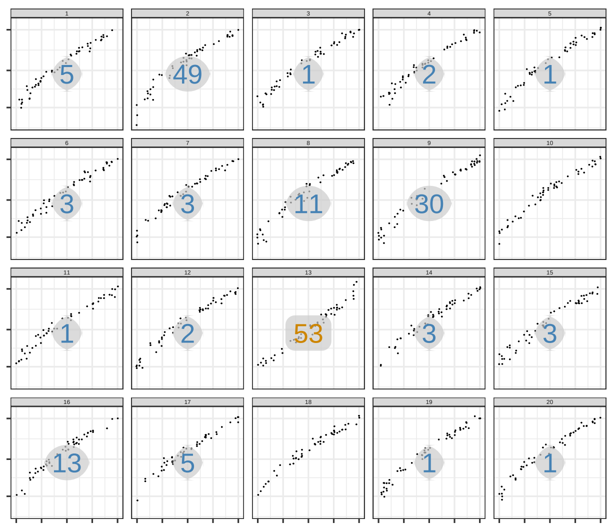
Target: Medium Curvature
Null: Low Curvature

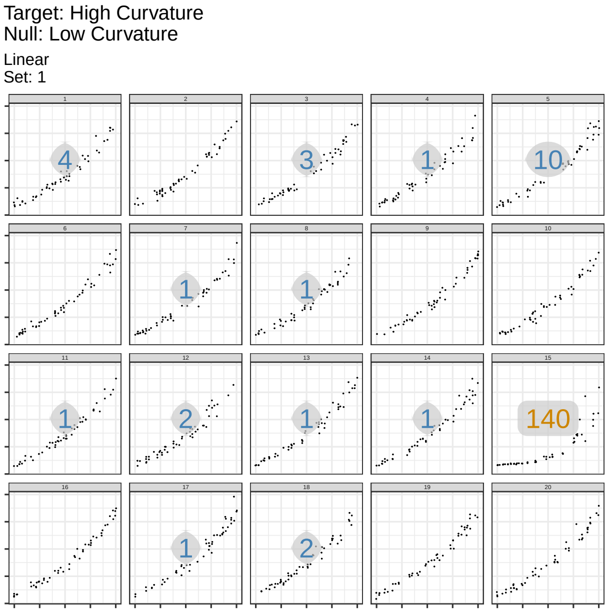
Linear
Set: 2



Target: Medium Curvature
Null: Low Curvature

Log
Set: 2

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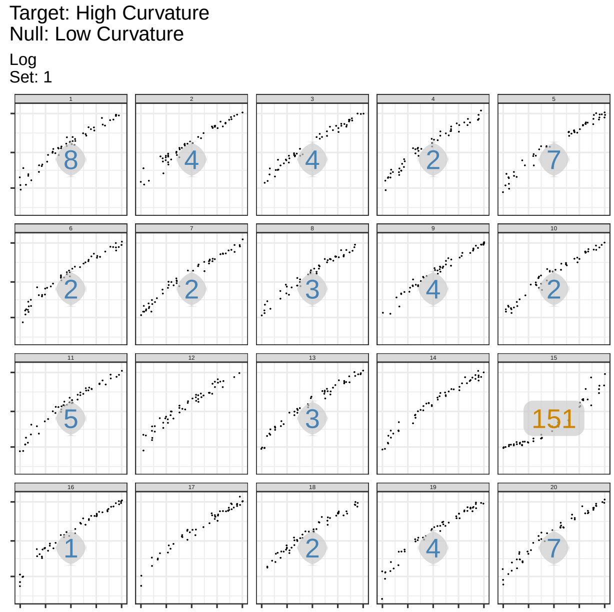
Linear
Set: 1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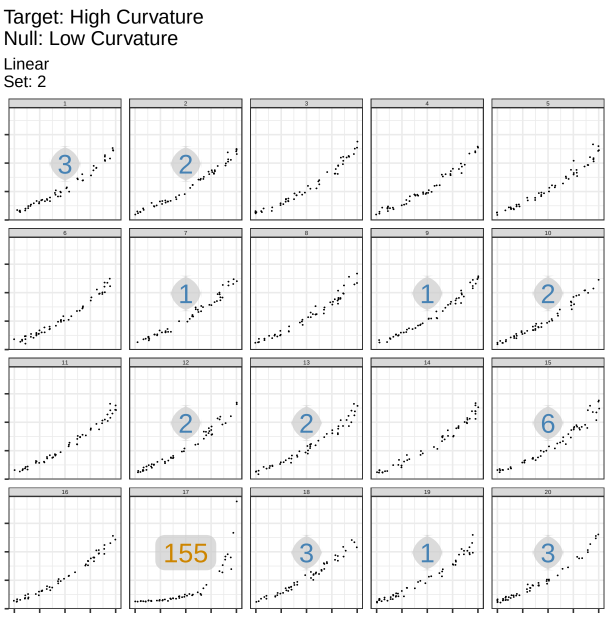
Null: Low Curvature
Log
Set: 1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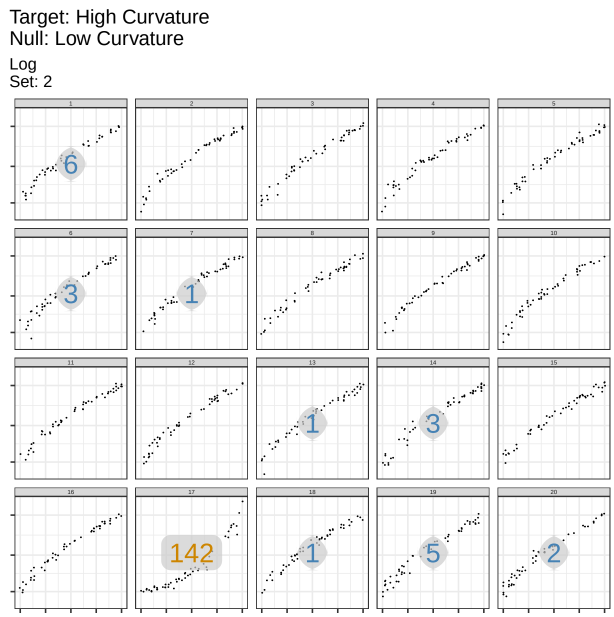
Linear
Set: 2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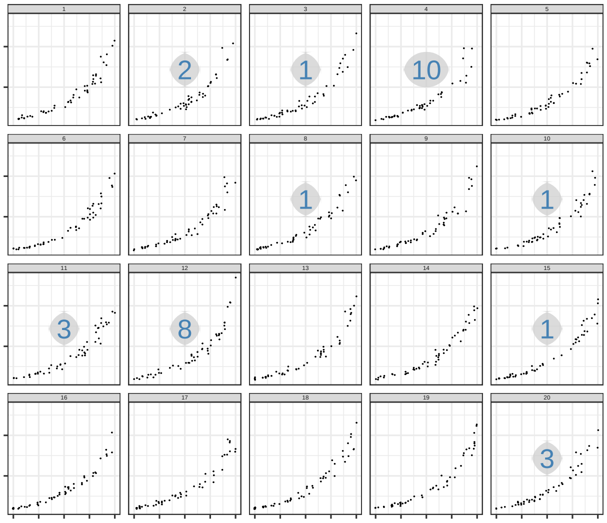
Log
Set: 2

C.2 Medium Curvature Null Background

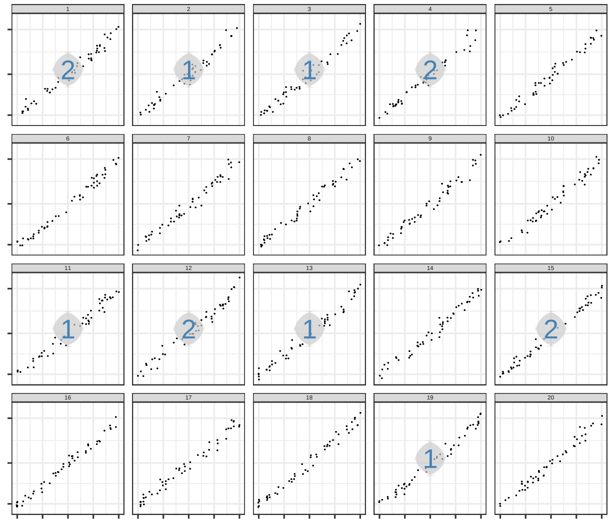
Target: Medium Curvature
Null: Medium Curvature

Linear
Set: 1



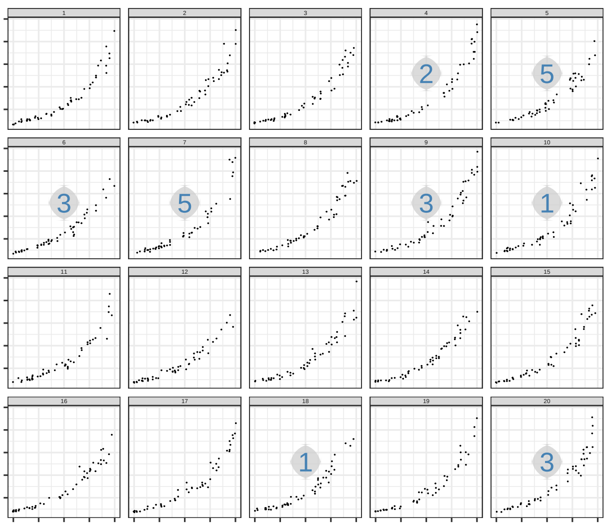
Target: Medium Curvature
Null: Medium Curvature

Log
Set: 1



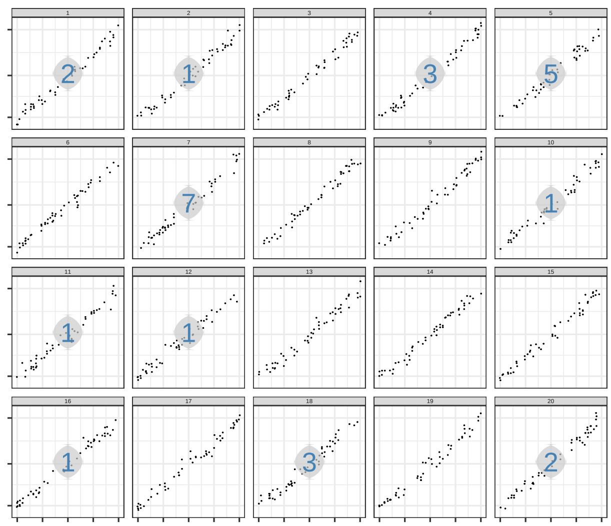
Target: Medium Curvature
Null: Medium Curvature

Linear
Set: 2



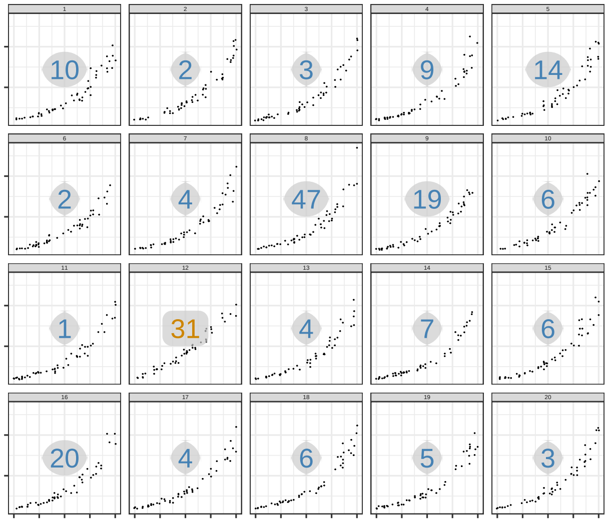
Target: Medium Curvature
Null: Medium Curvature

Log
Set: 2



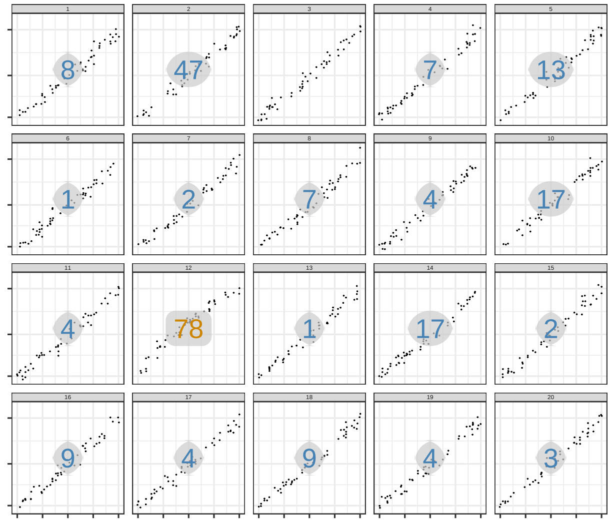
Target: Low Curvature
Null: Medium Curvature

Linear
Set: 1



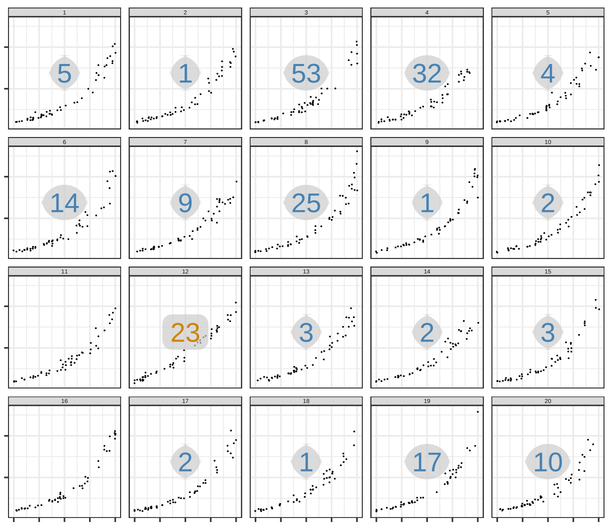
Target: Low Curvature
Null: Medium Curvature

Log
Set: 1



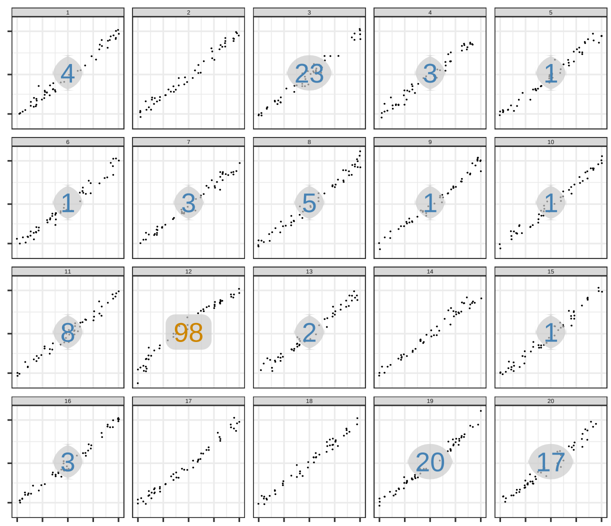
Target: Low Curvature
Null: Medium Curvature

Linear
Set: 2



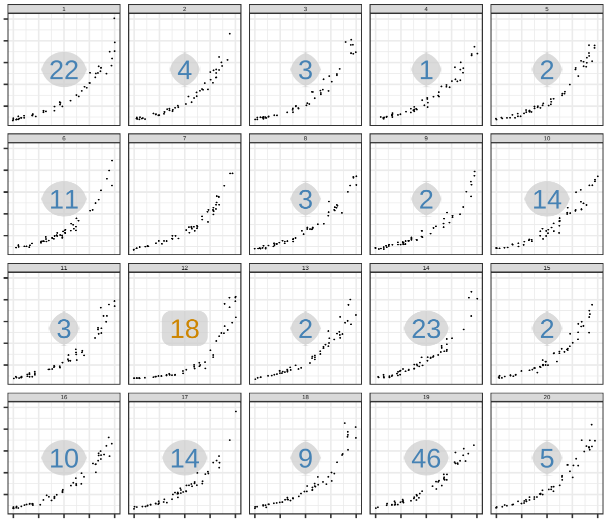
Target: Low Curvature
Null: Medium Curvature

Log
Set: 2



Target: High Curvature
Null: Medium Curvature

Linear
Set: 1

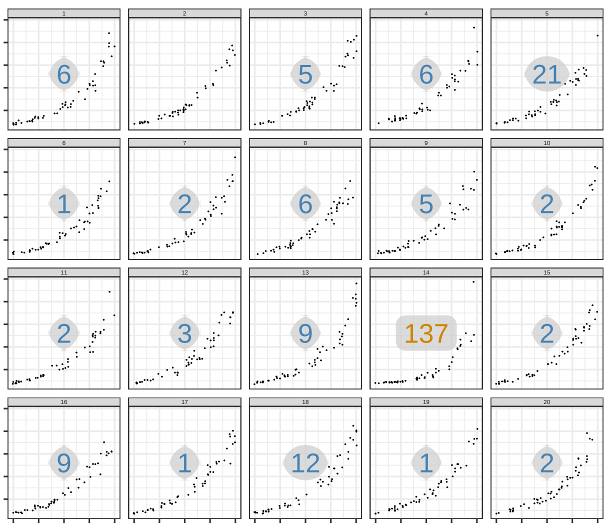


Target: High Curvature
Null: Medium Curvature

Log
Set: 1

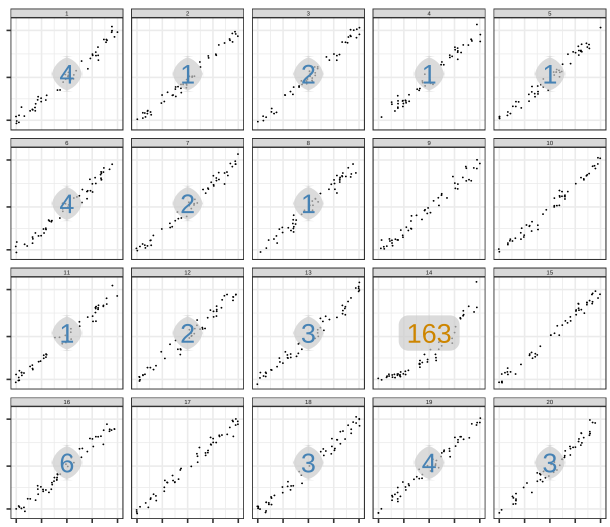
Target: High Curvature
Null: Medium Curvature

Linear
Set: 2



Target: High Curvature
Null: Medium Curvature

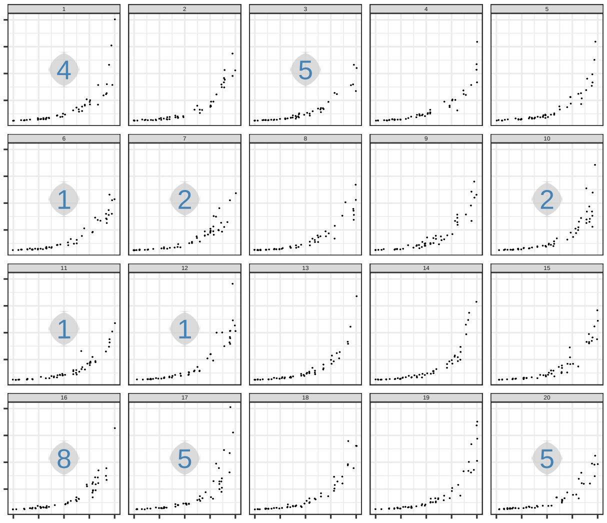
Log
Set: 2



C.3 High Curvature Null Background

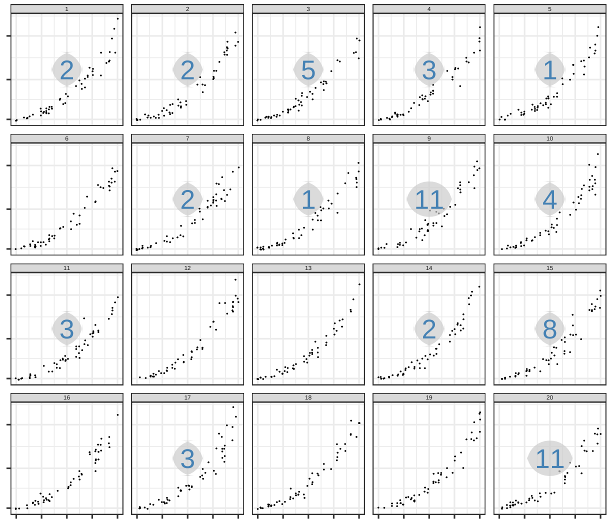
Target: High Curvature
Null: High Curvature

Linear
Set: 1



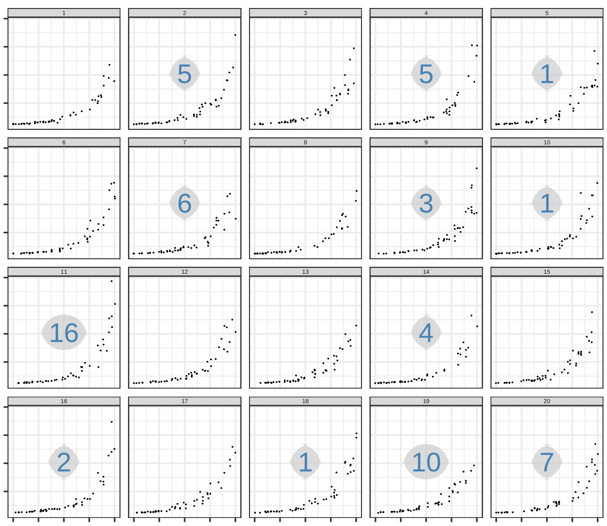
Target: High Curvature
Null: High Curvature

Log
Set: 1



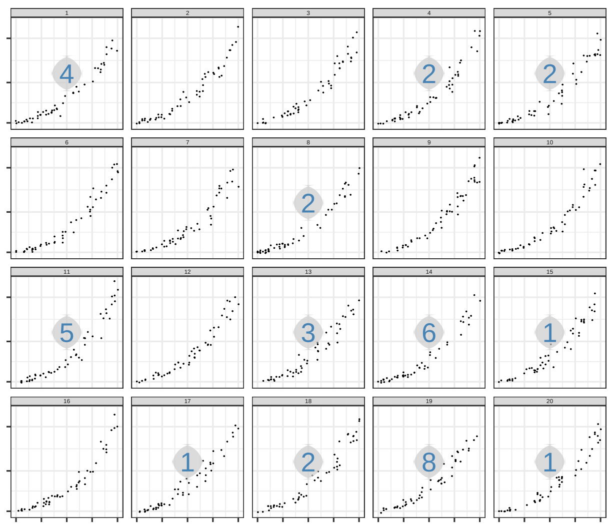
Target: High Curvature
Null: High Curvature

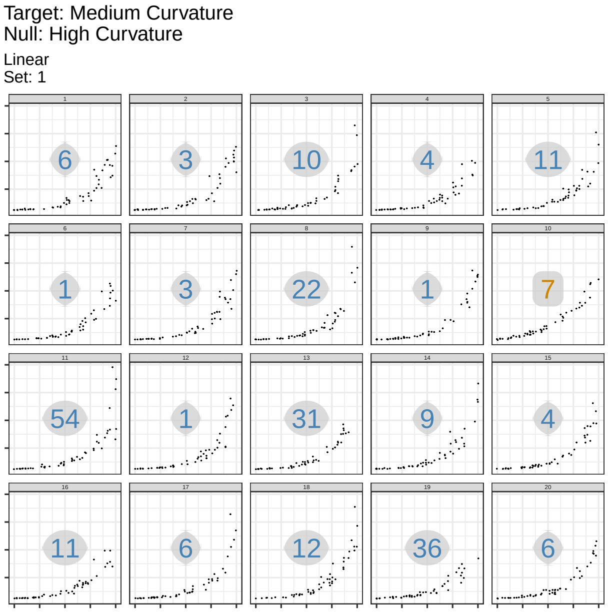
Linear
Set: 2



Target: High Curvature
Null: High Curvature

Log
Set: 2




Target: Medium Curvature
Null: High Curvature
Linear
Set: 1
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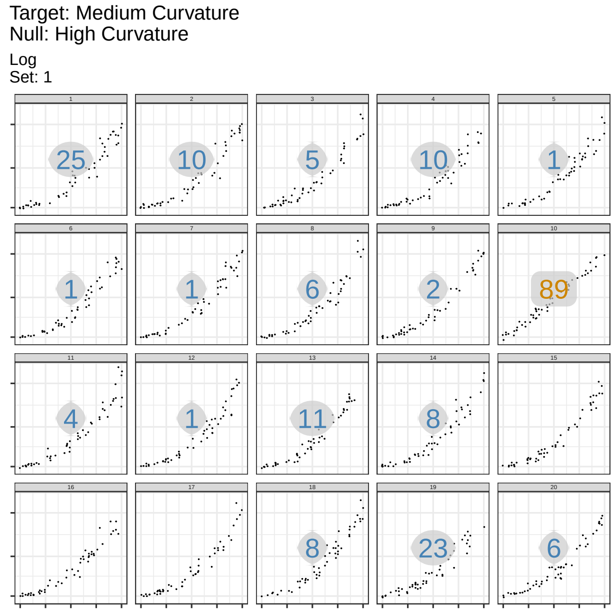
6
12
36
6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Null: High Curvature
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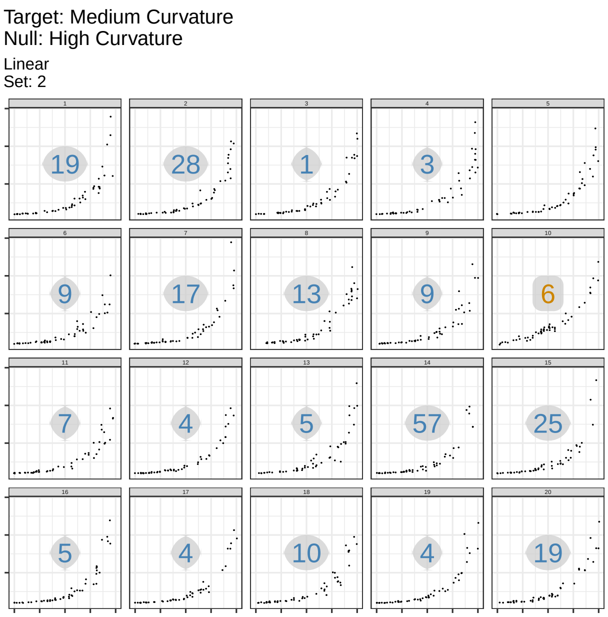
8
8
23
6


Target: Medium Curvature
Null: High Curvature
Linear
Set: 2
19
28
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4
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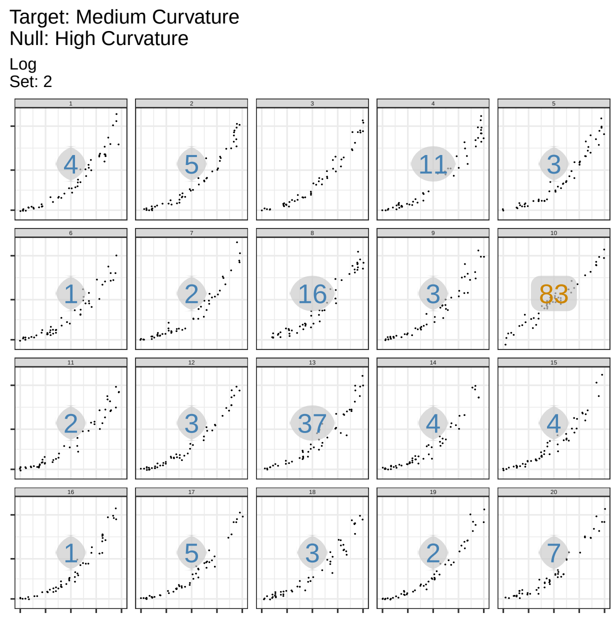
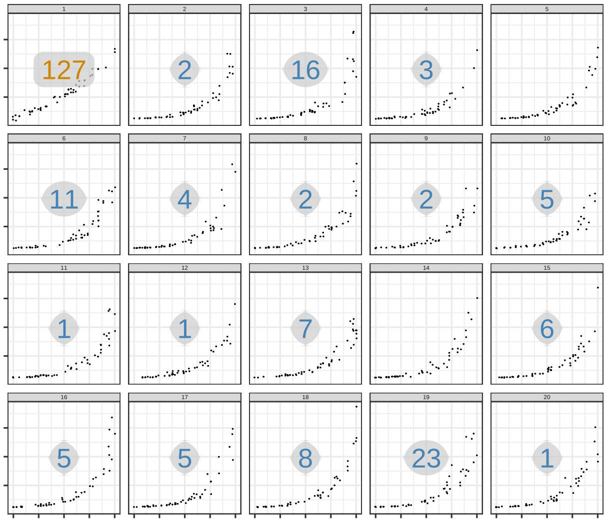
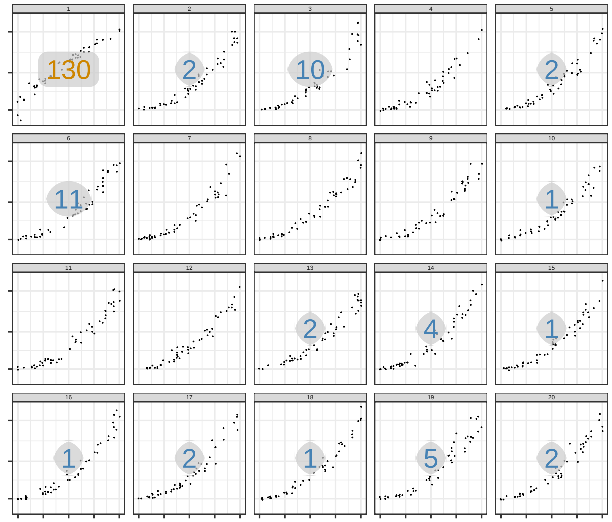
25
5
4
10
4
19


Target: Medium Curvature
Null: High Curvature
Log
Set: 2
4
5
11
3
1
2
16
3
83
2
3
37
4
4
1
5
3
2
7

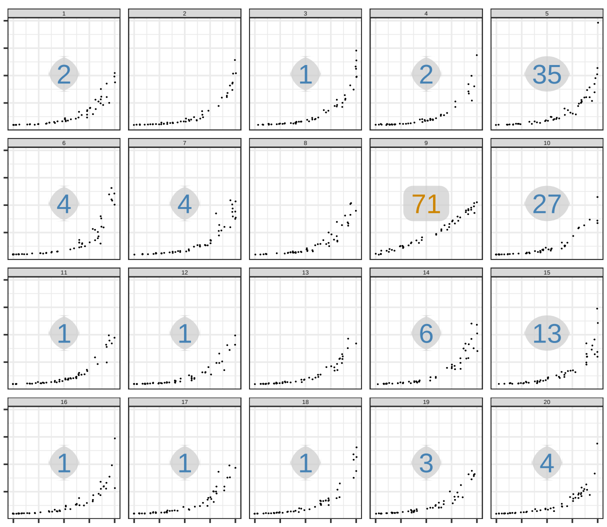
Target: Low Curvature
Null: High Curvature
Linear
Set: 1



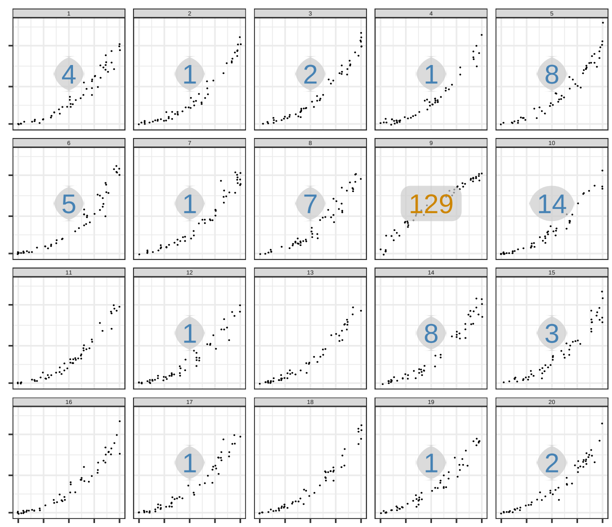
Target: Low Curvature
Null: High Curvature
Log
Set: 1



Target: Low Curvature
Null: High Curvature
Linear
Set: 2



Target: Low Curvature
Null: High Curvature
Log
Set: 2



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